

mw-atlas

Towards the three-dimensional shape of the Milky Way in Gas and Dust

Fabian Polnitzky

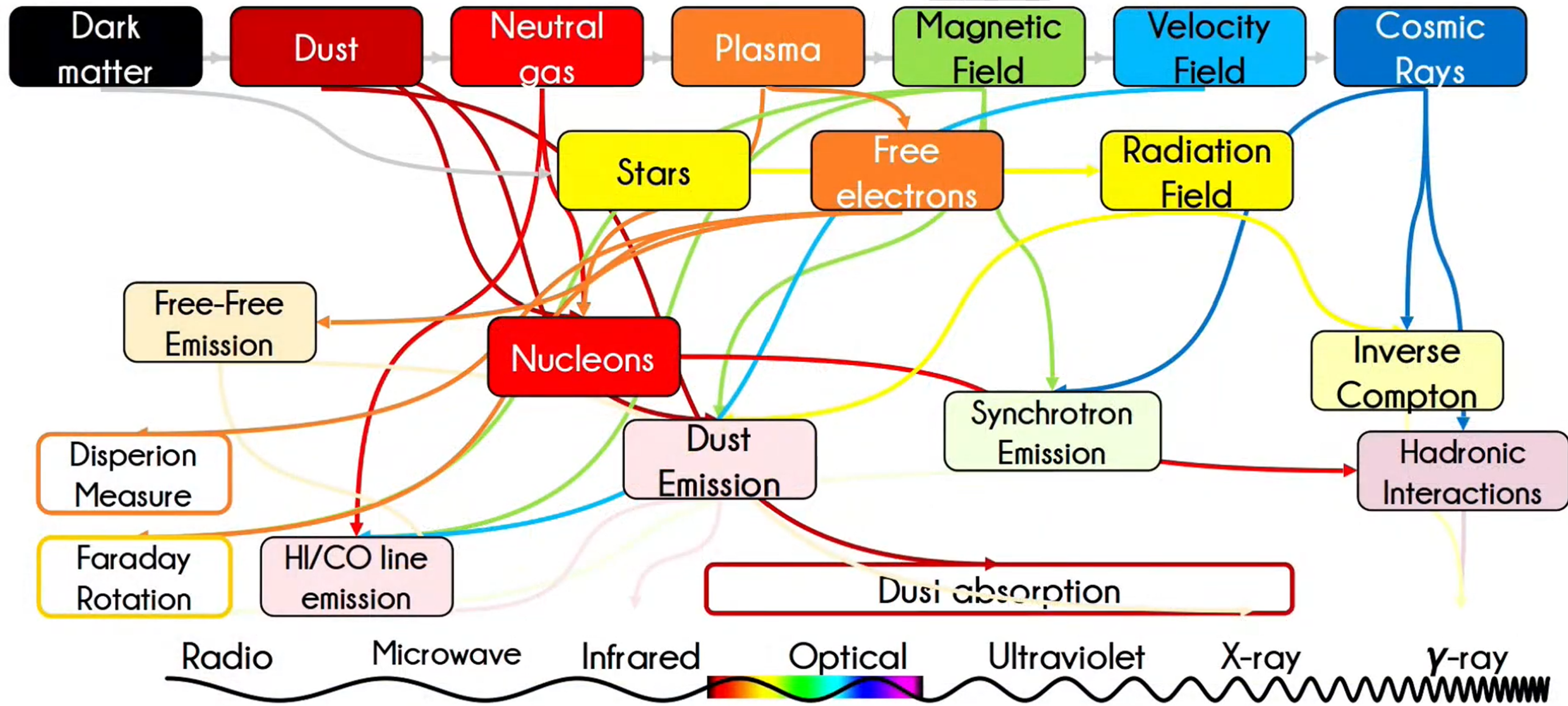
Together With

Laurin Söding, Philipp Frank, Mareike Berkner, Richa Halder, Matthias Hübl,

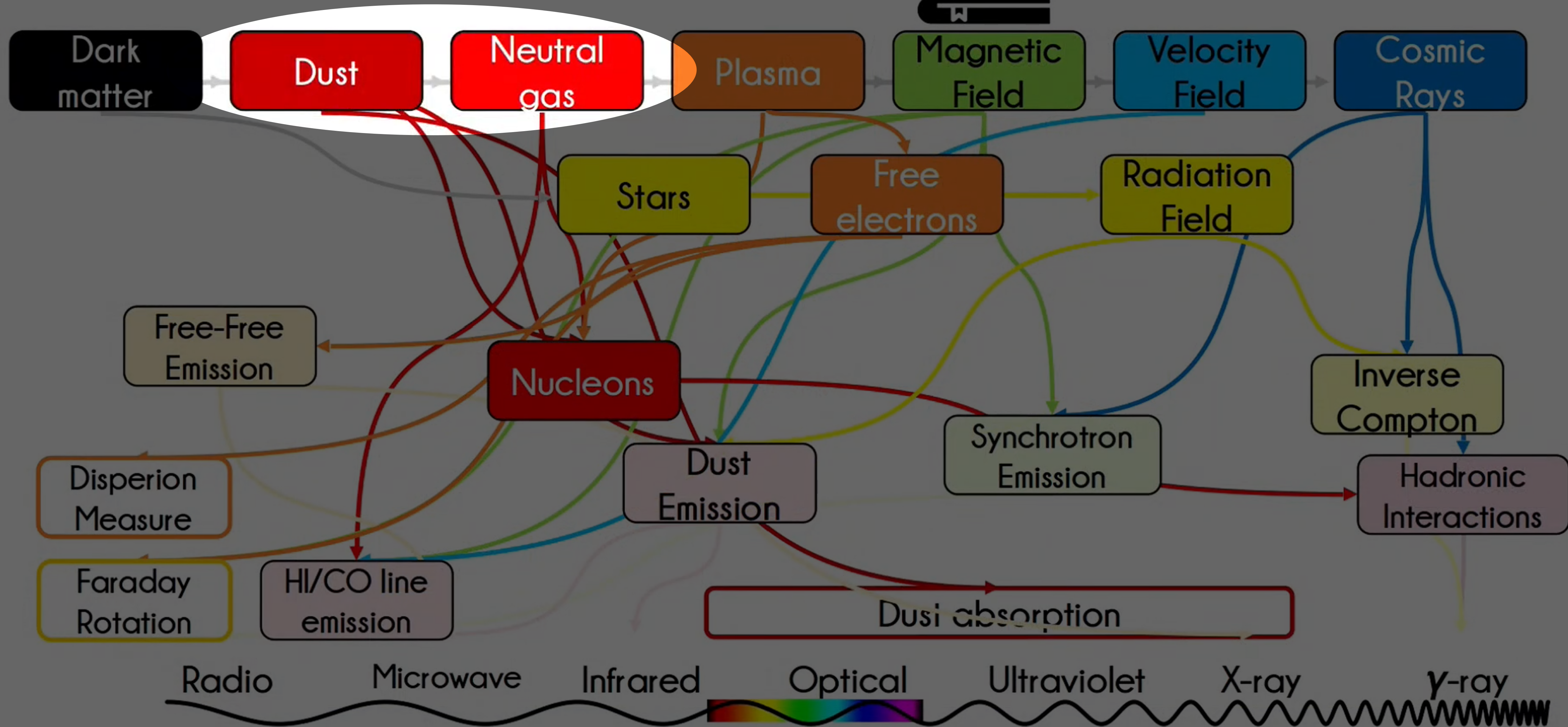
Philipp Mertsch + mw-atlas Team

Star Formation Fields, September 21st 2026

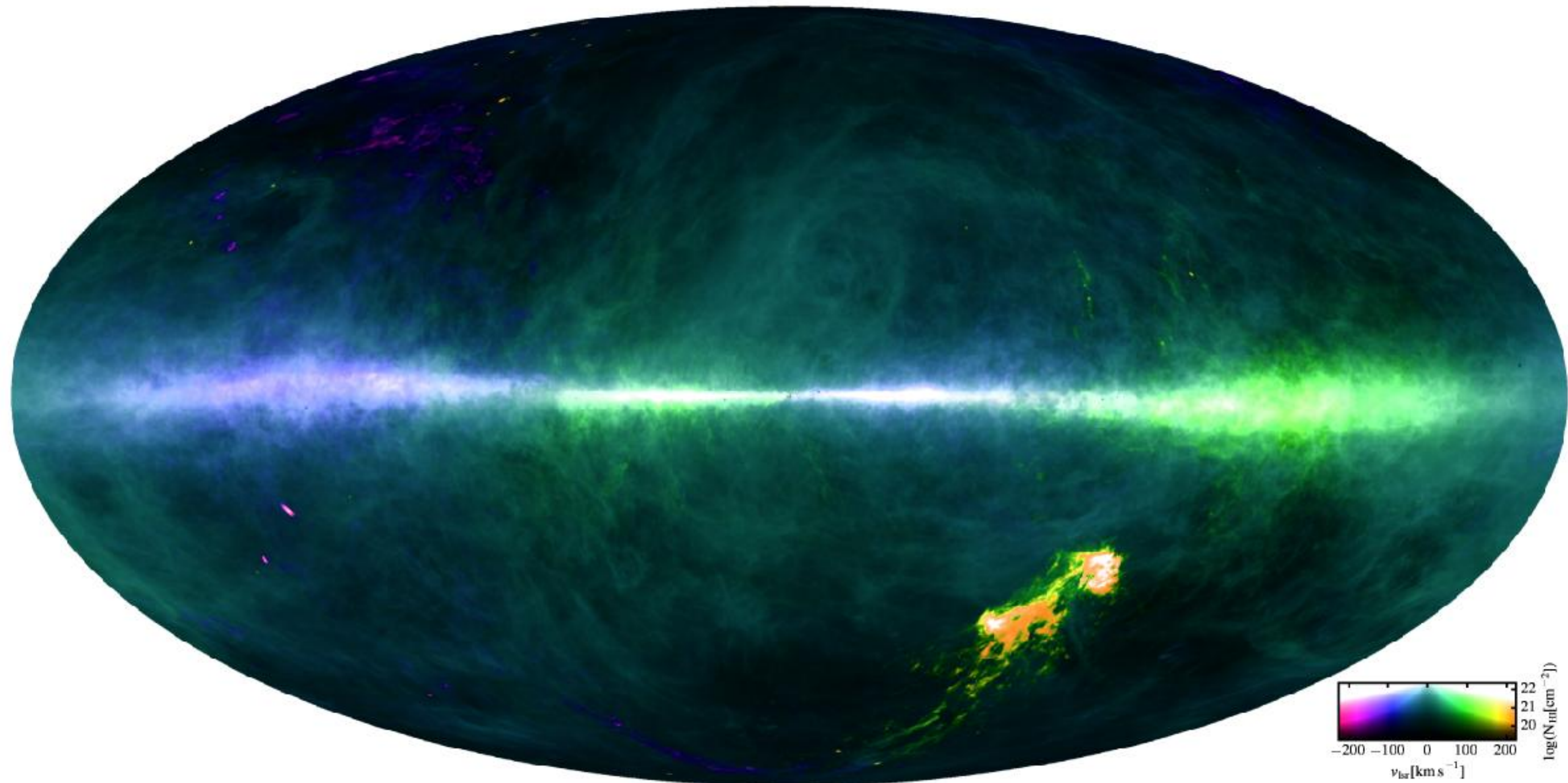
Milky Way Atlas



Milky Way Atlas



Milky Way in Neutral Hydrogen - HI4PI

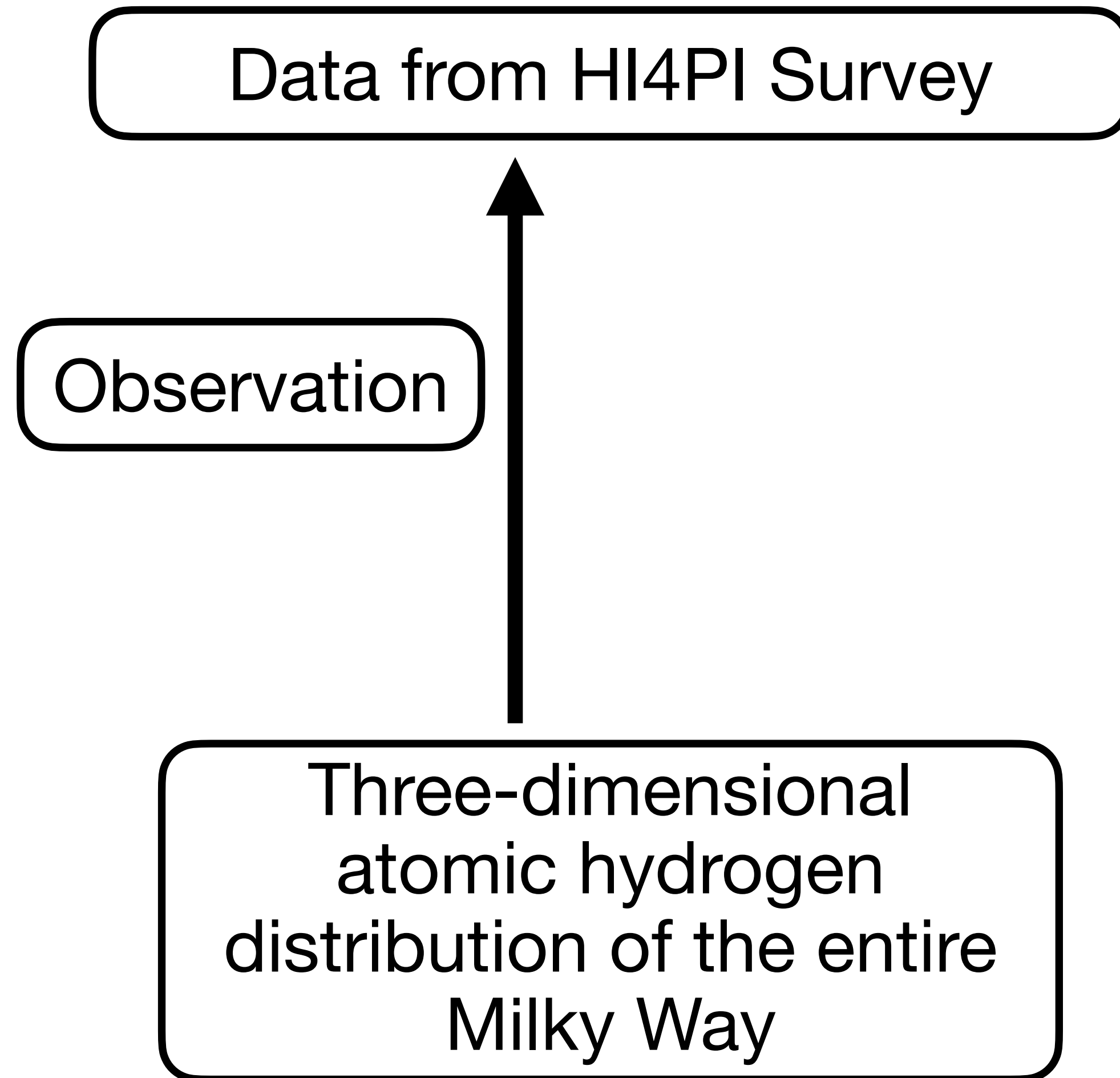


Söding et al. 2024 - Hydrogen in the Galaxy

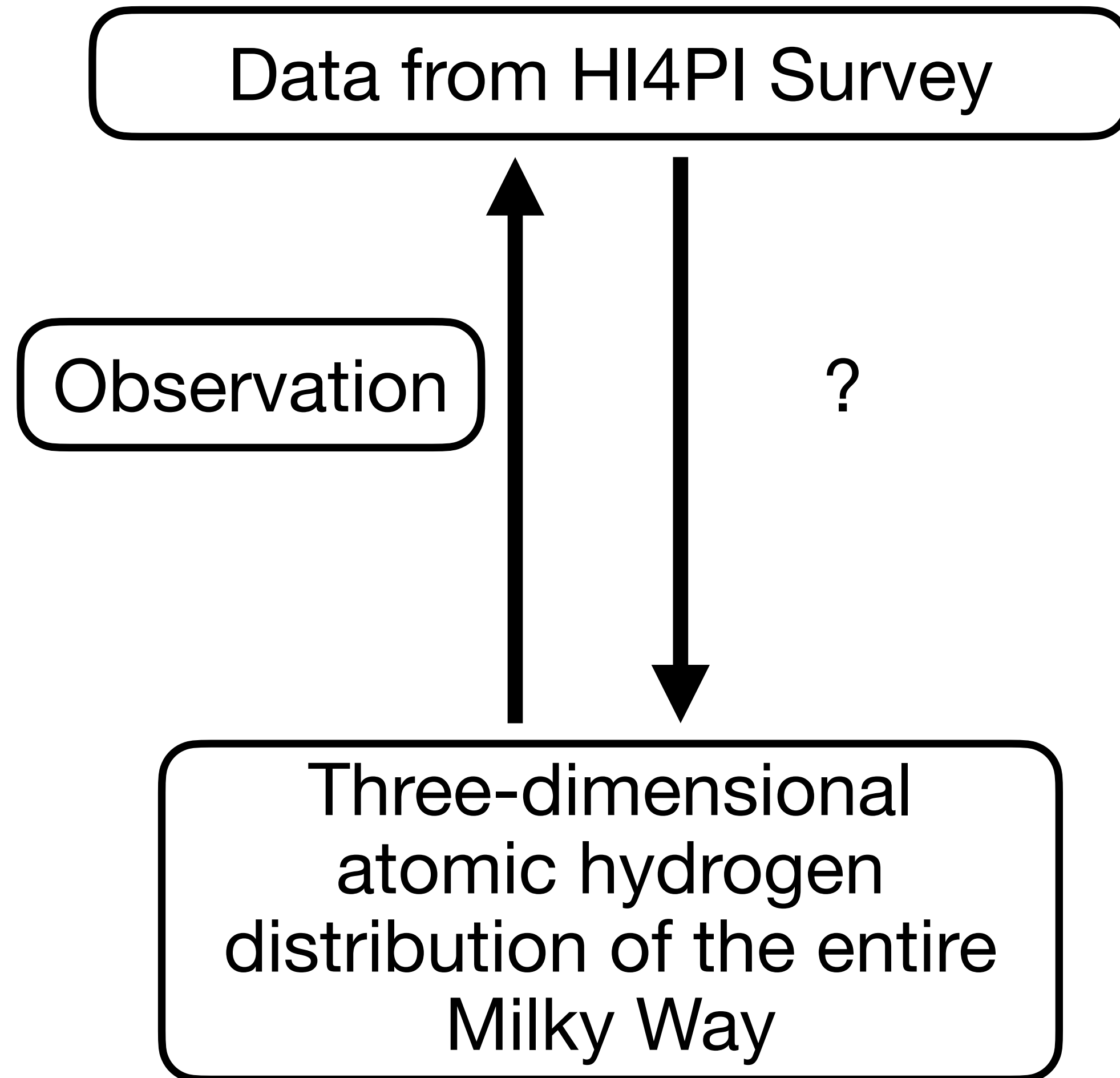
Söding et al. 2024 - Hydrogen in the Galaxy

Three-dimensional
atomic hydrogen
distribution of the entire
Milky Way

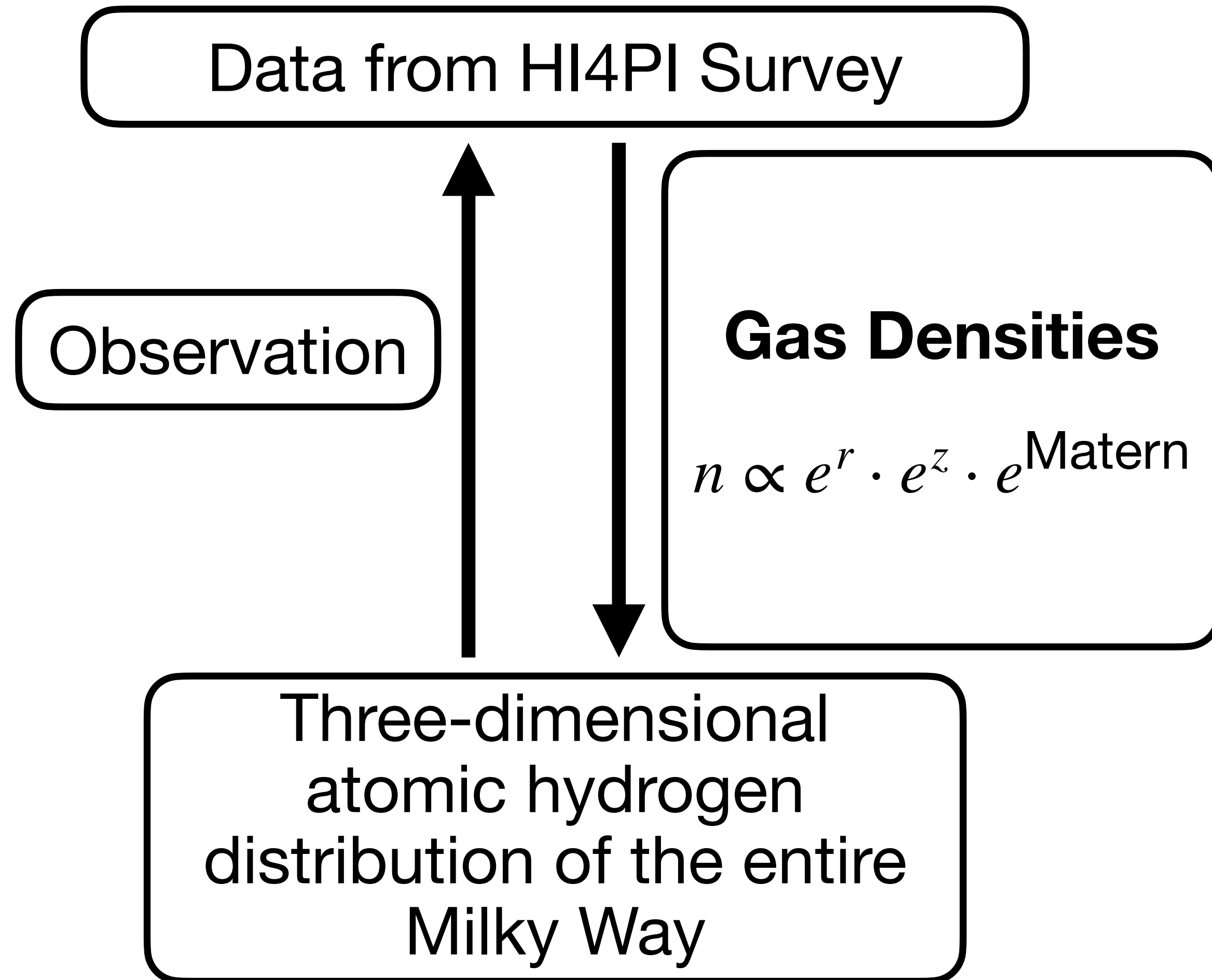
Söding et al. 2024 - Hydrogen in the Galaxy



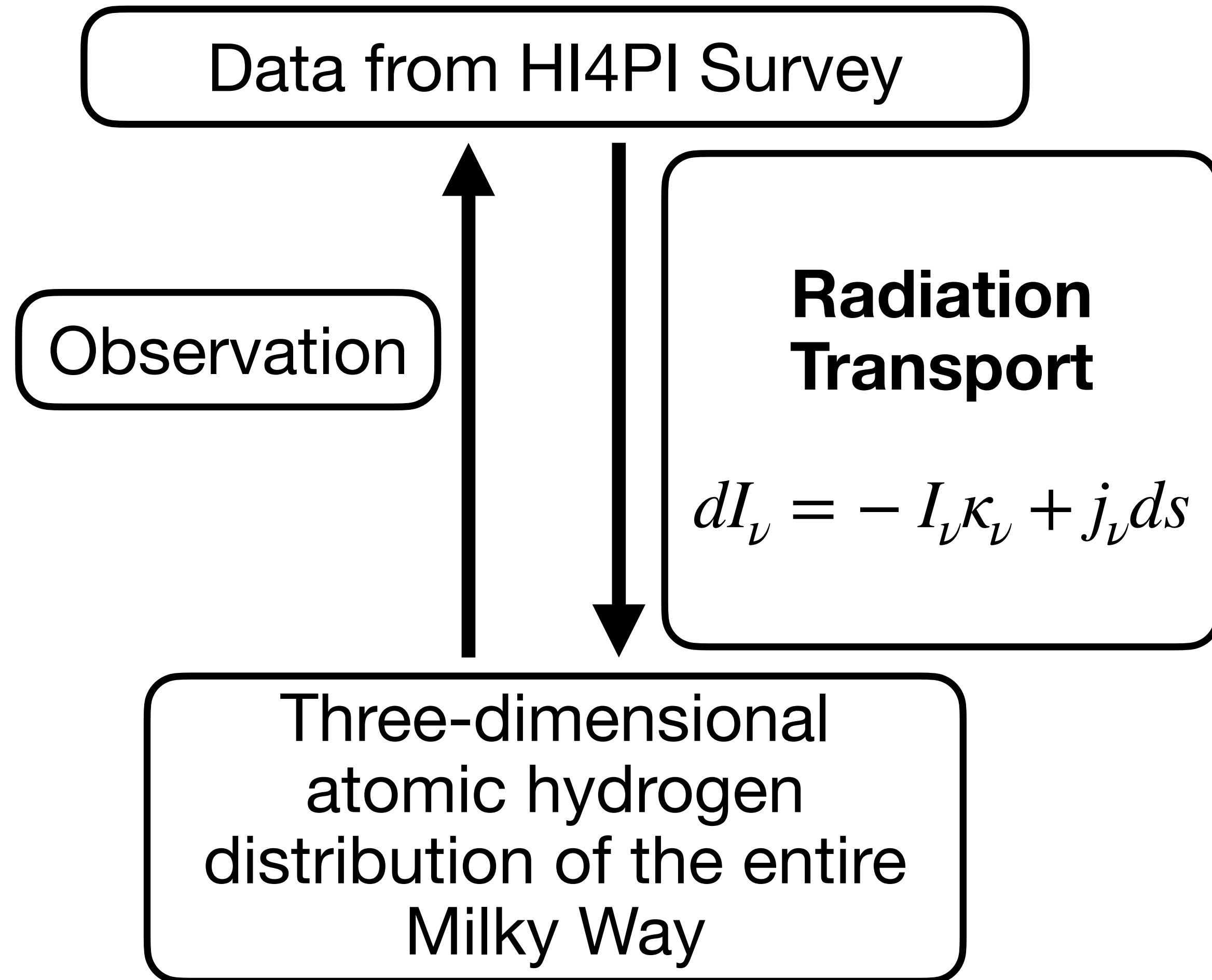
Söding et al. 2024 - Hydrogen in the Galaxy



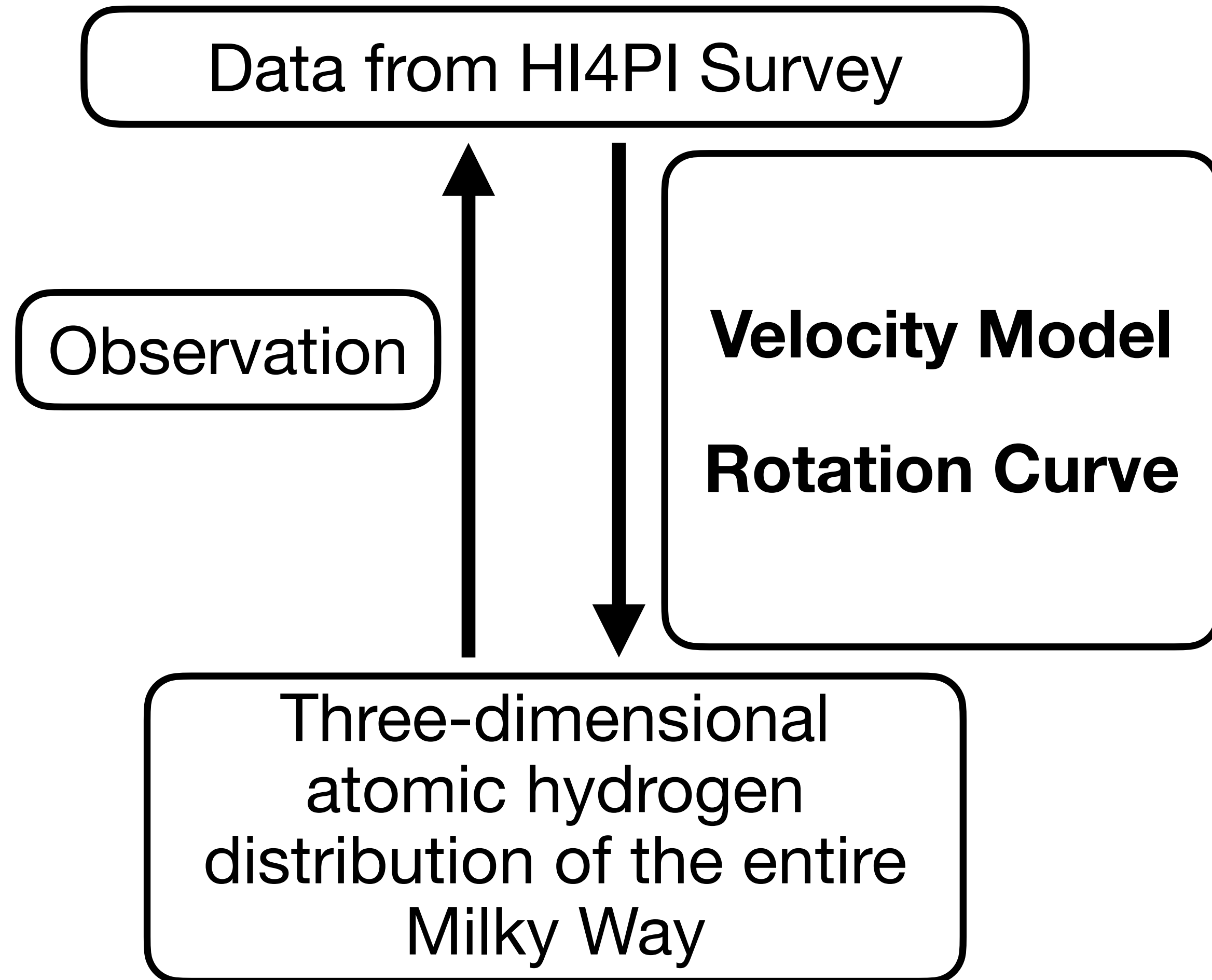
Söding et al. 2024 - Hydrogen in the Galaxy



Söding et al. 2024 - Hydrogen in the Galaxy

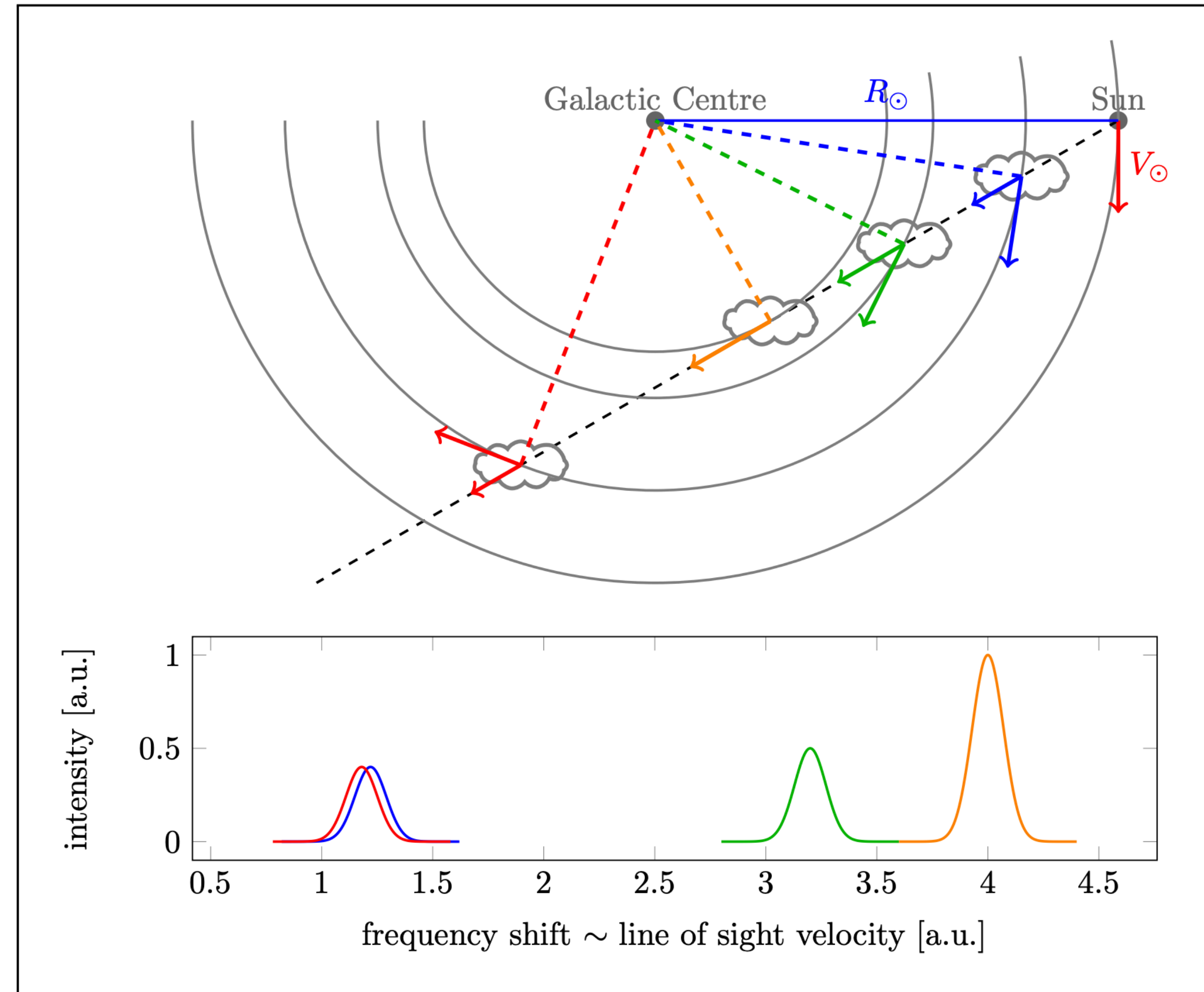
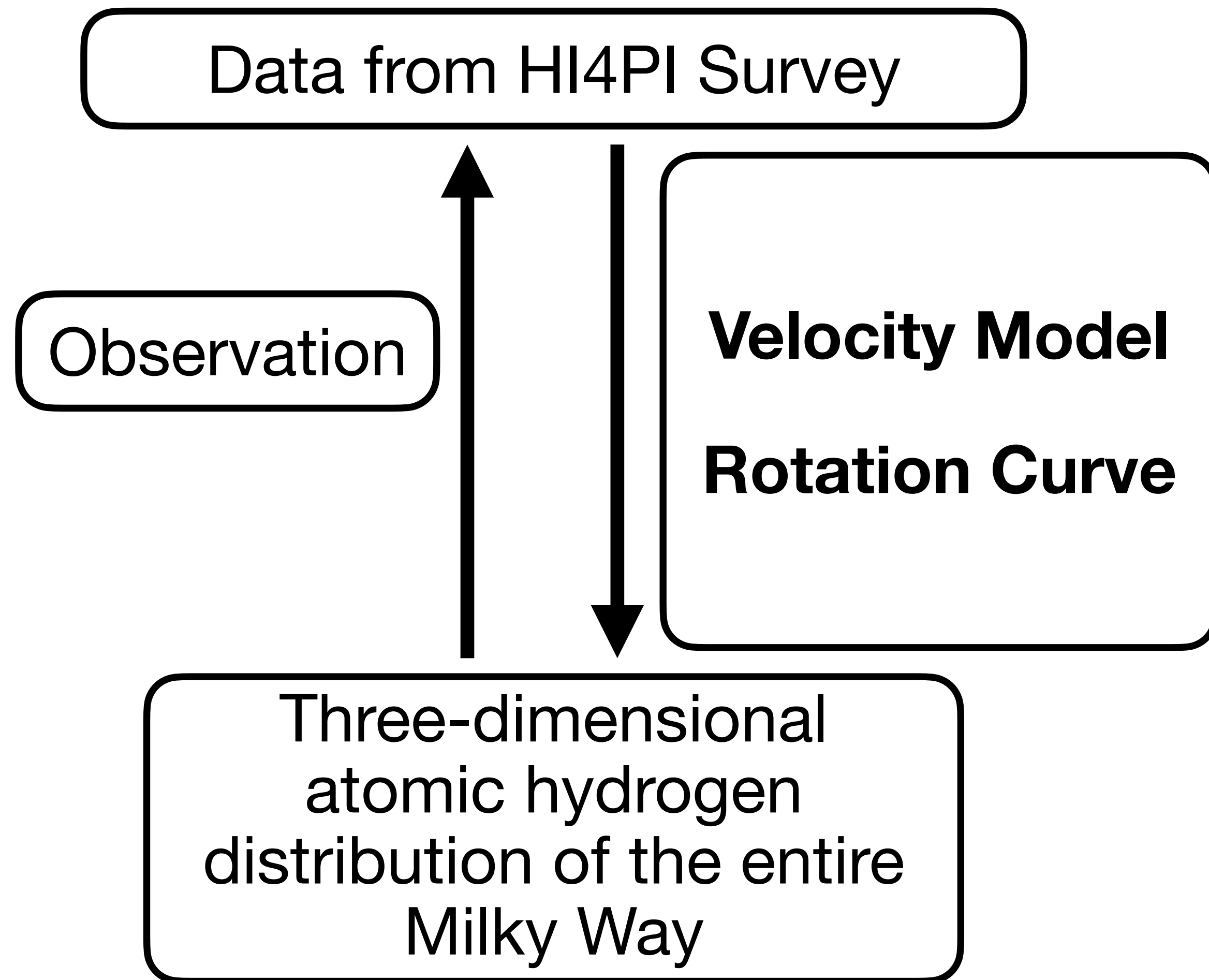


Söding et al. 2024 - Hydrogen in the Galaxy

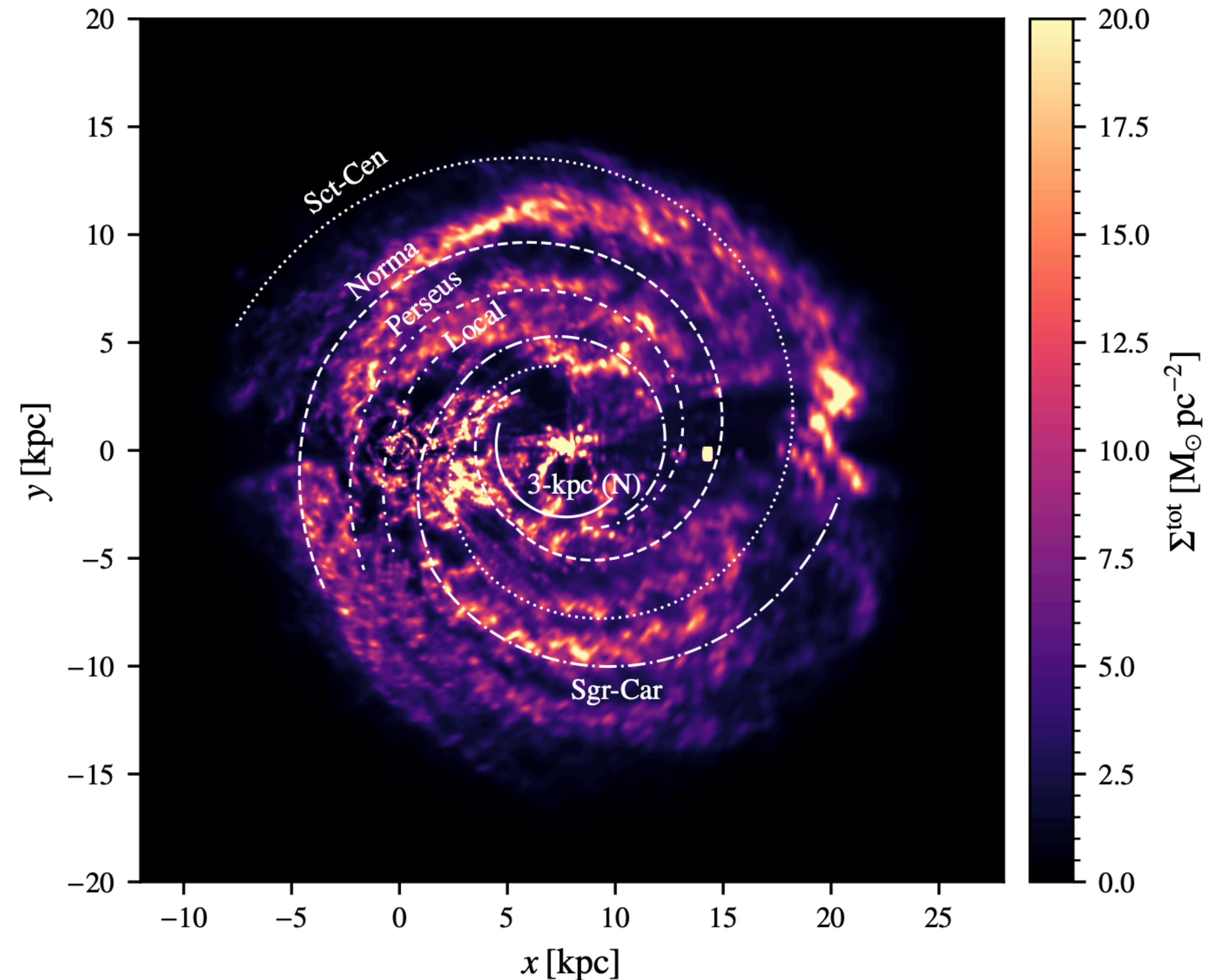
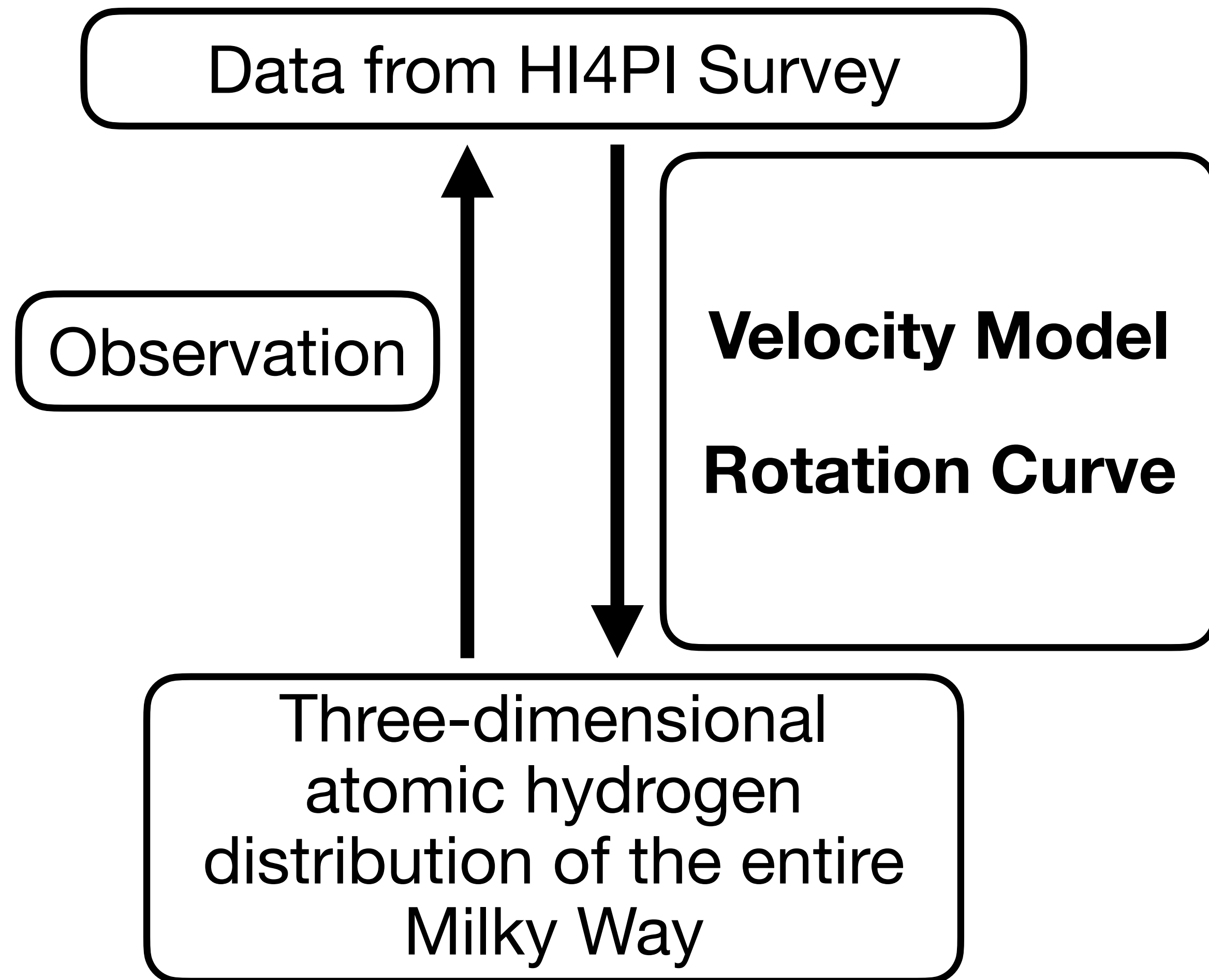


Söding et al. 2024 - Hydrogen in the Galaxy

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Söding et al. 2024 - Hydrogen in the Galaxy

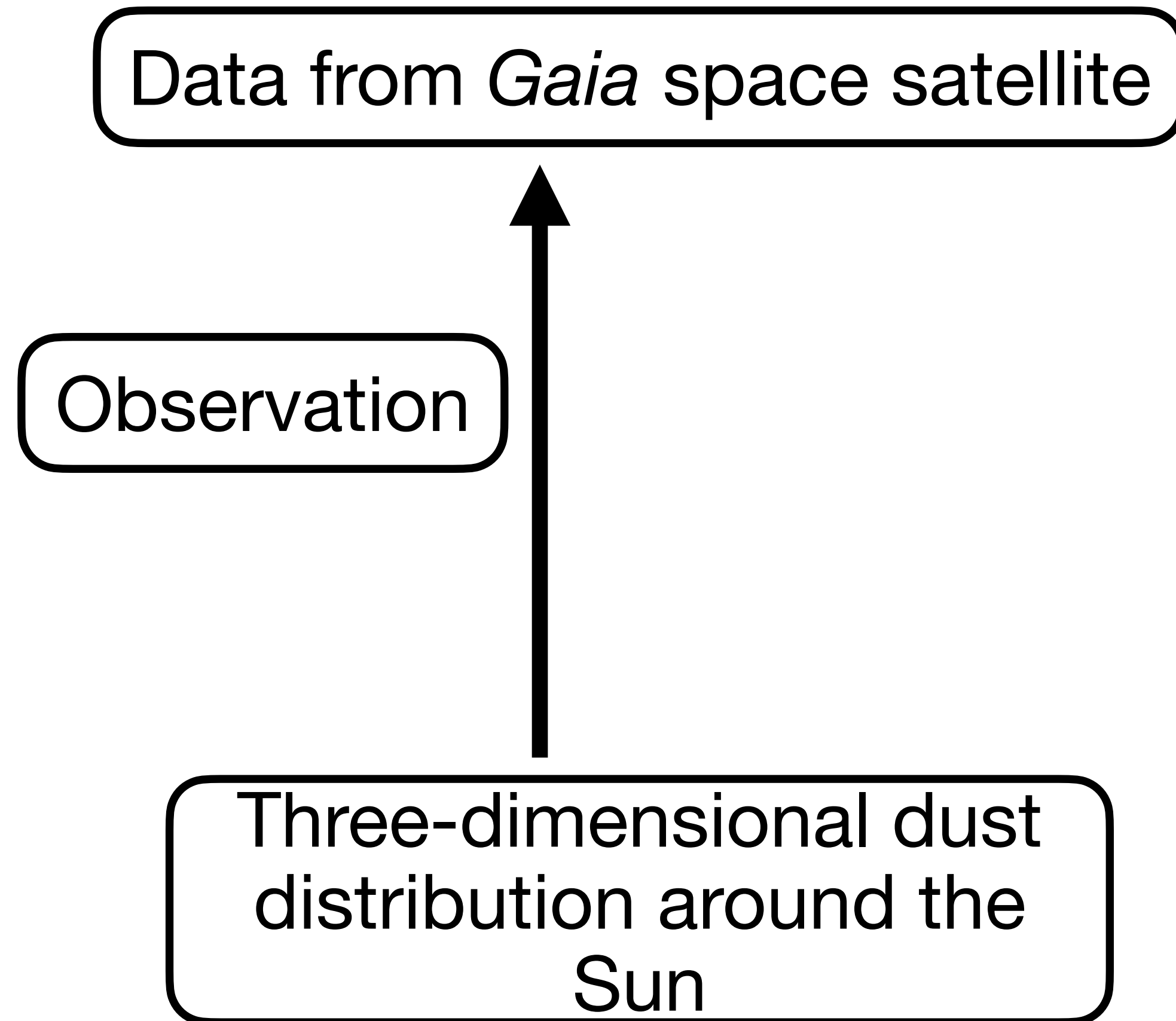


Edenhofer et al. 2024 - Dust around the Sun

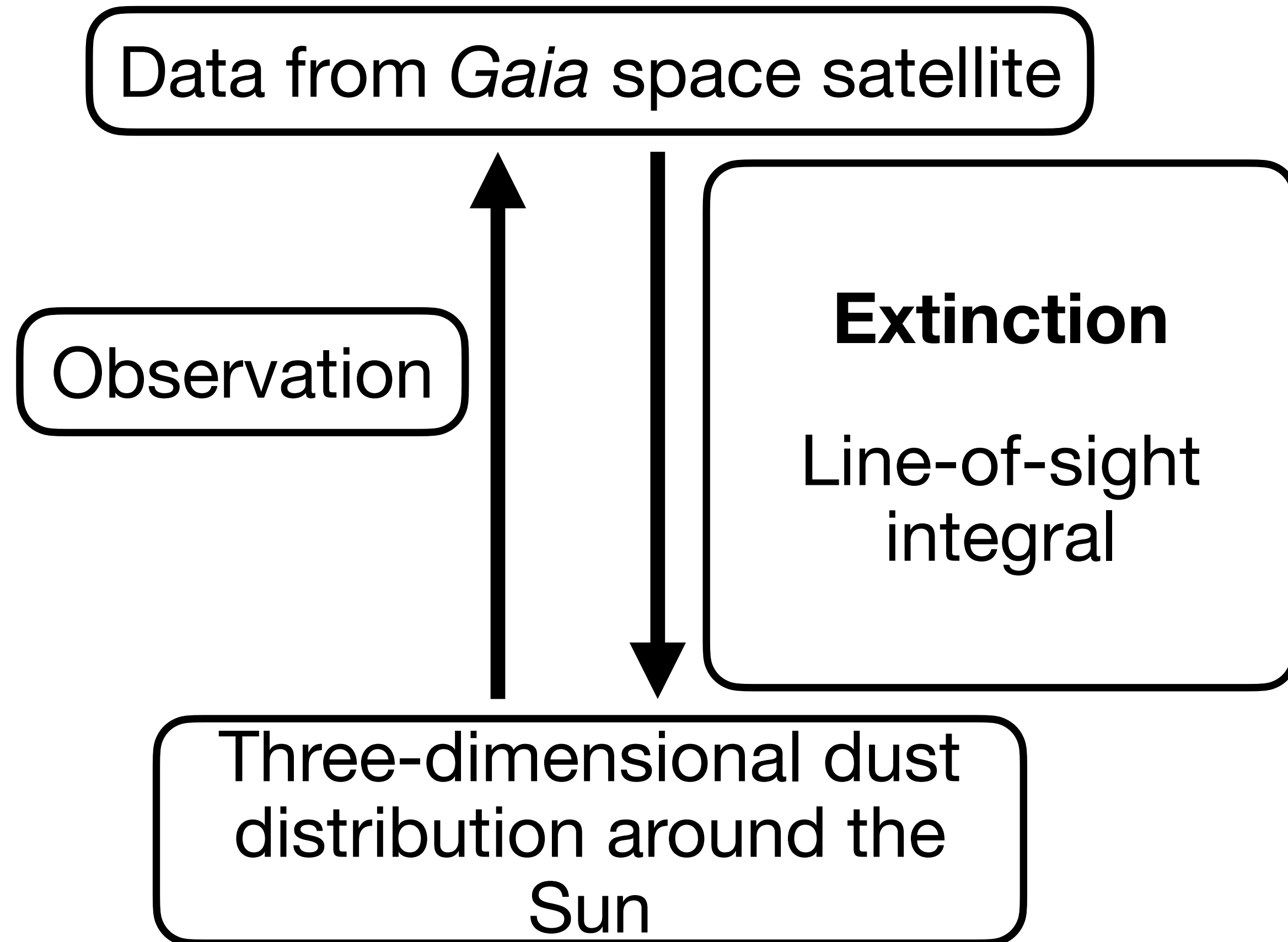
Edenhofer et al. 2024 - Dust around the Sun

Three-dimensional dust
distribution around the
Sun

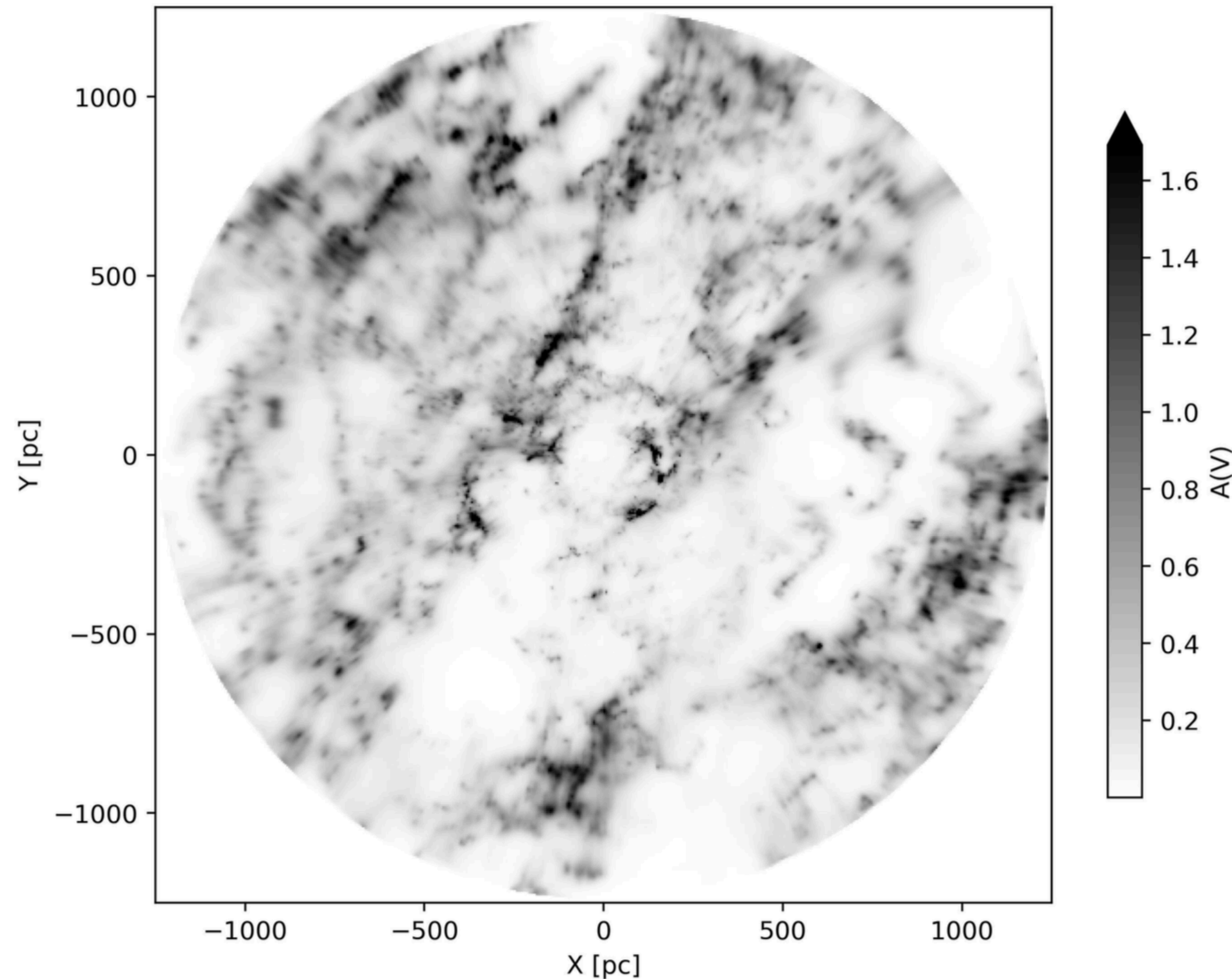
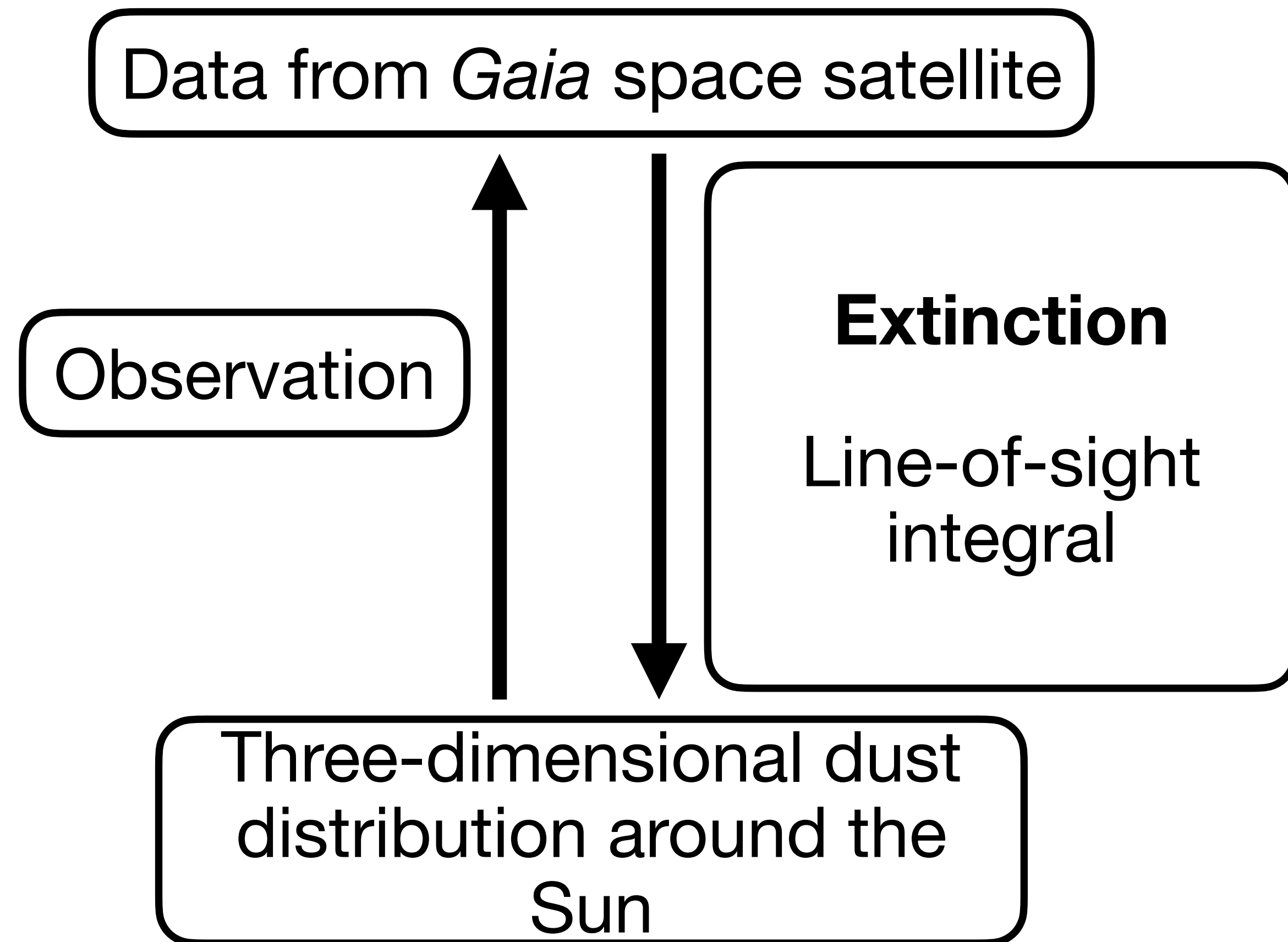
Edenhofer et al. 2024 - Dust around the Sun



Edenhofer et al. 2024 - Dust around the Sun



Edenhofer et al. 2024 - Dust around the Sun



Combining the two components

- Known correlation of hydrogen gas and interstellar dust

Combining the two components

- Known correlation of hydrogen gas and interstellar dust
- Map hydrogen gas and interstellar dust simultaneously

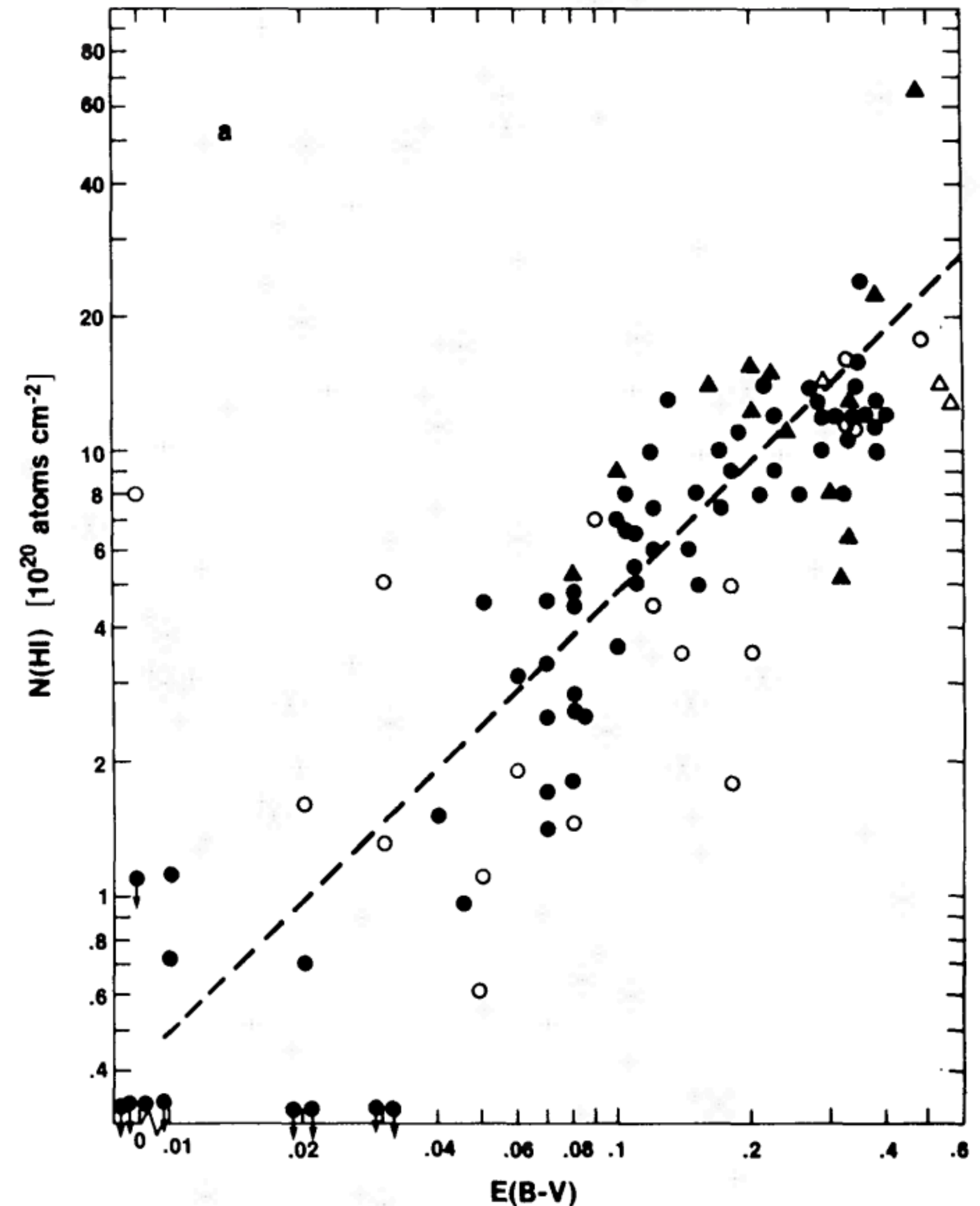
Combining the two components

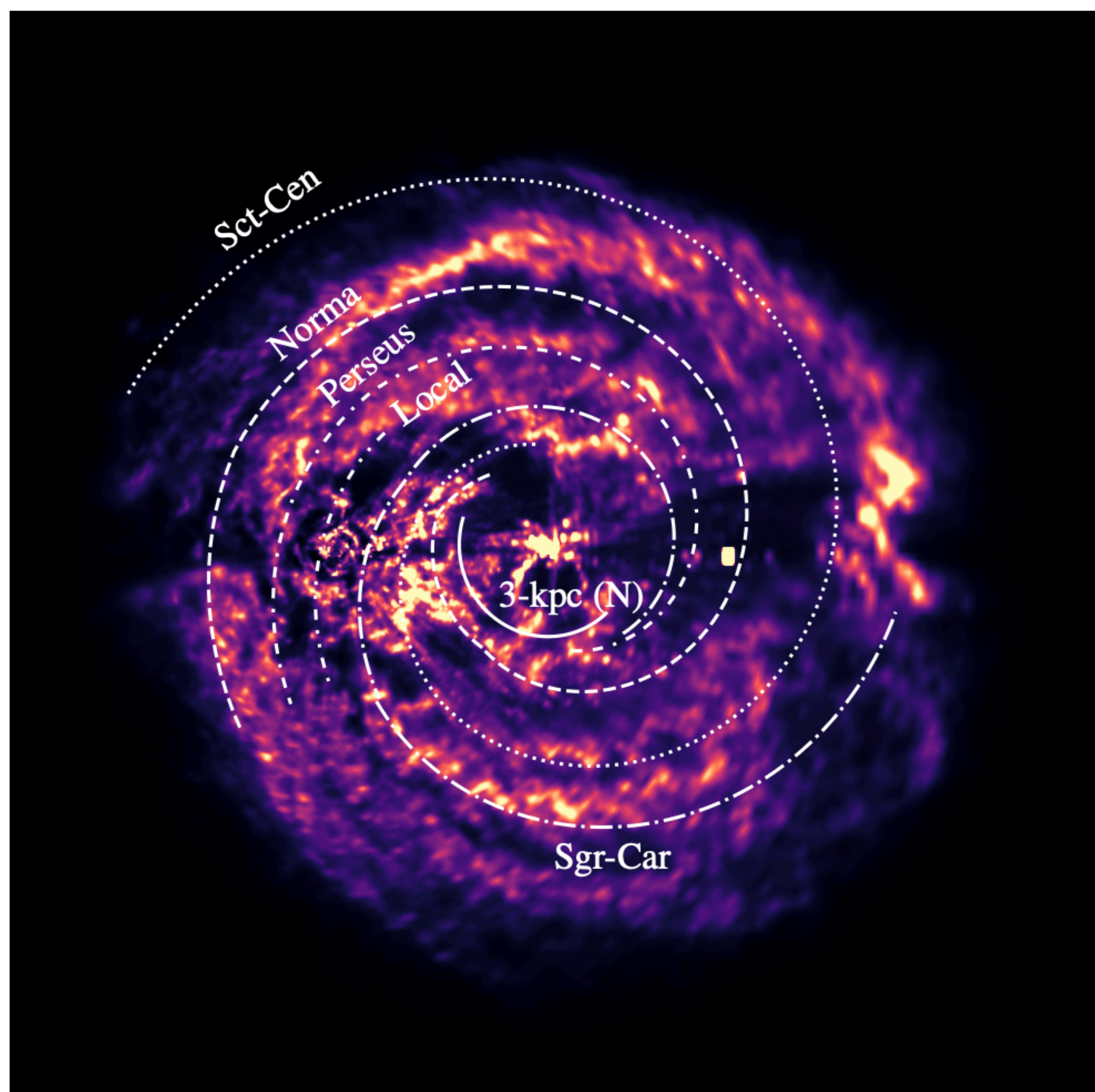
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- Map hydrogen gas and interstellar dust simultaneously
- Let both influence the other one

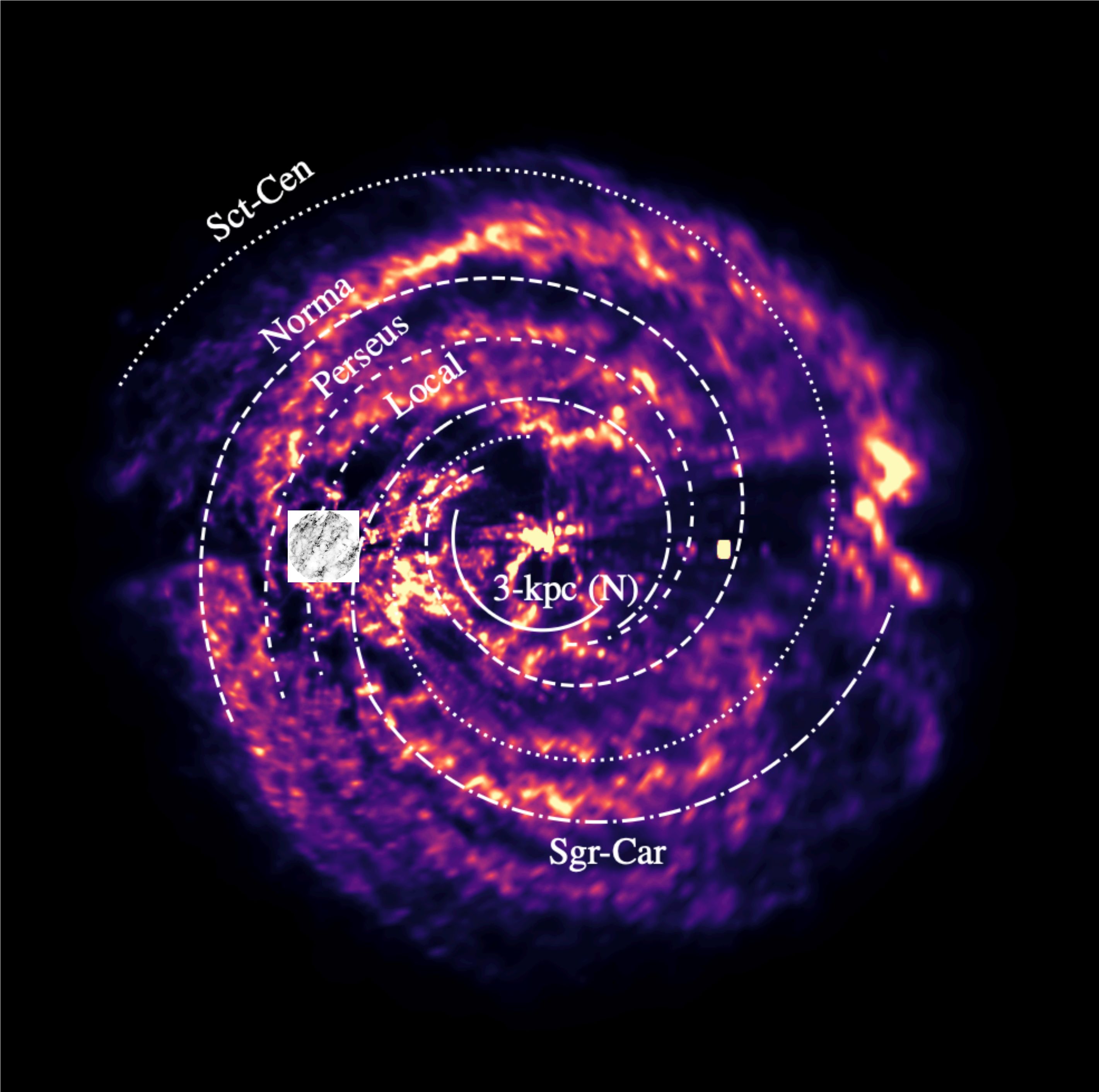
Combining the two components

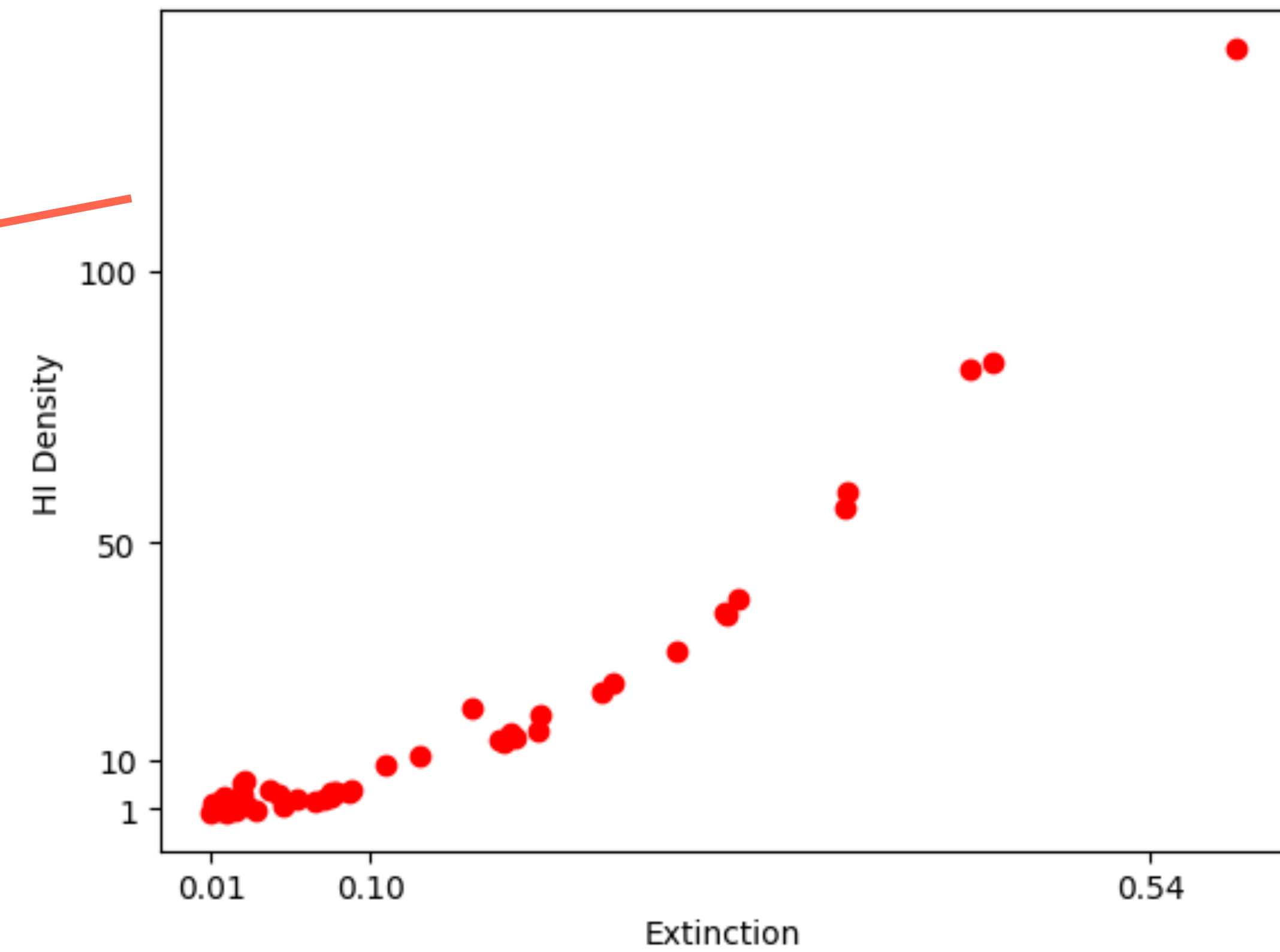
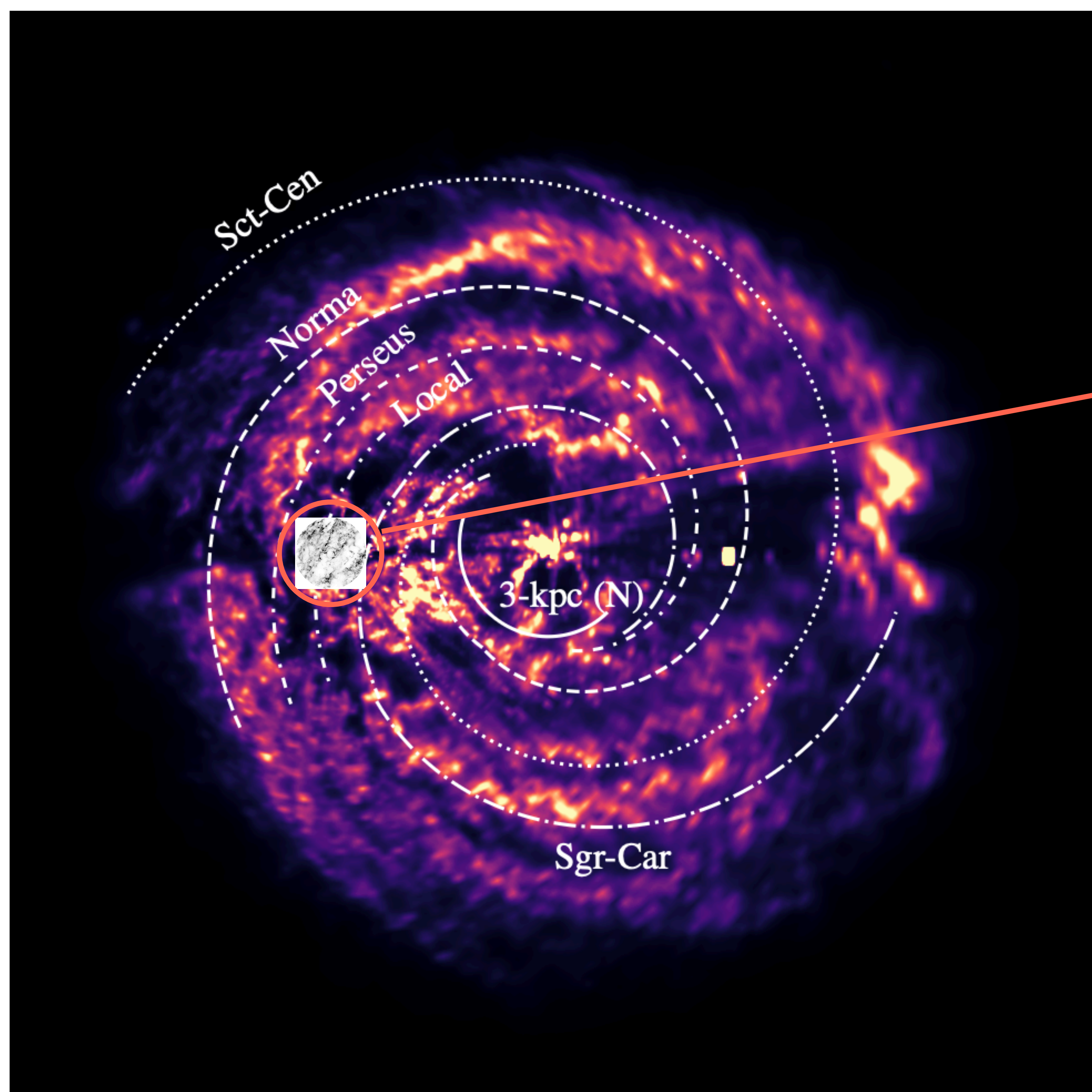
Bohlin, Savage & Drake 1978

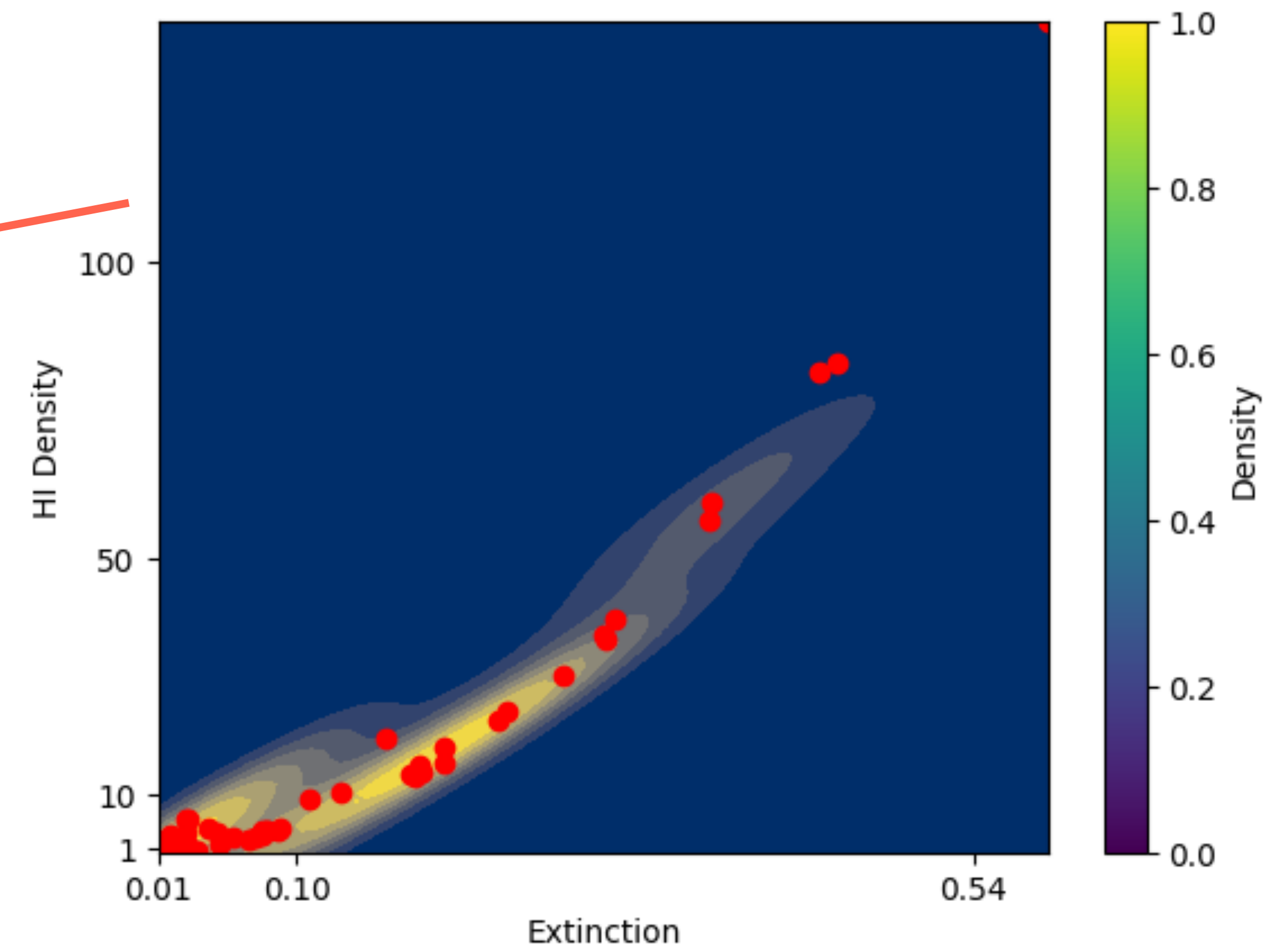
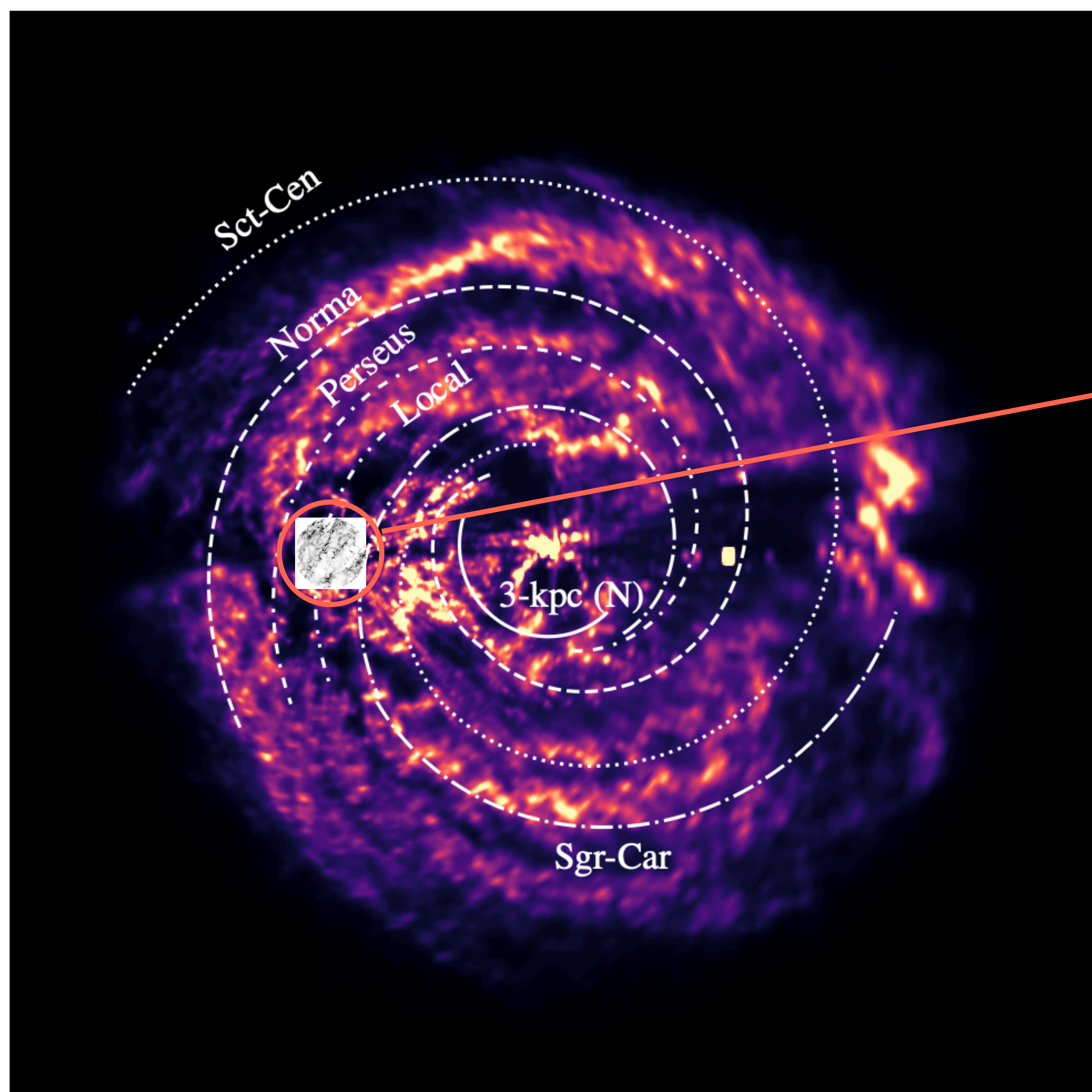
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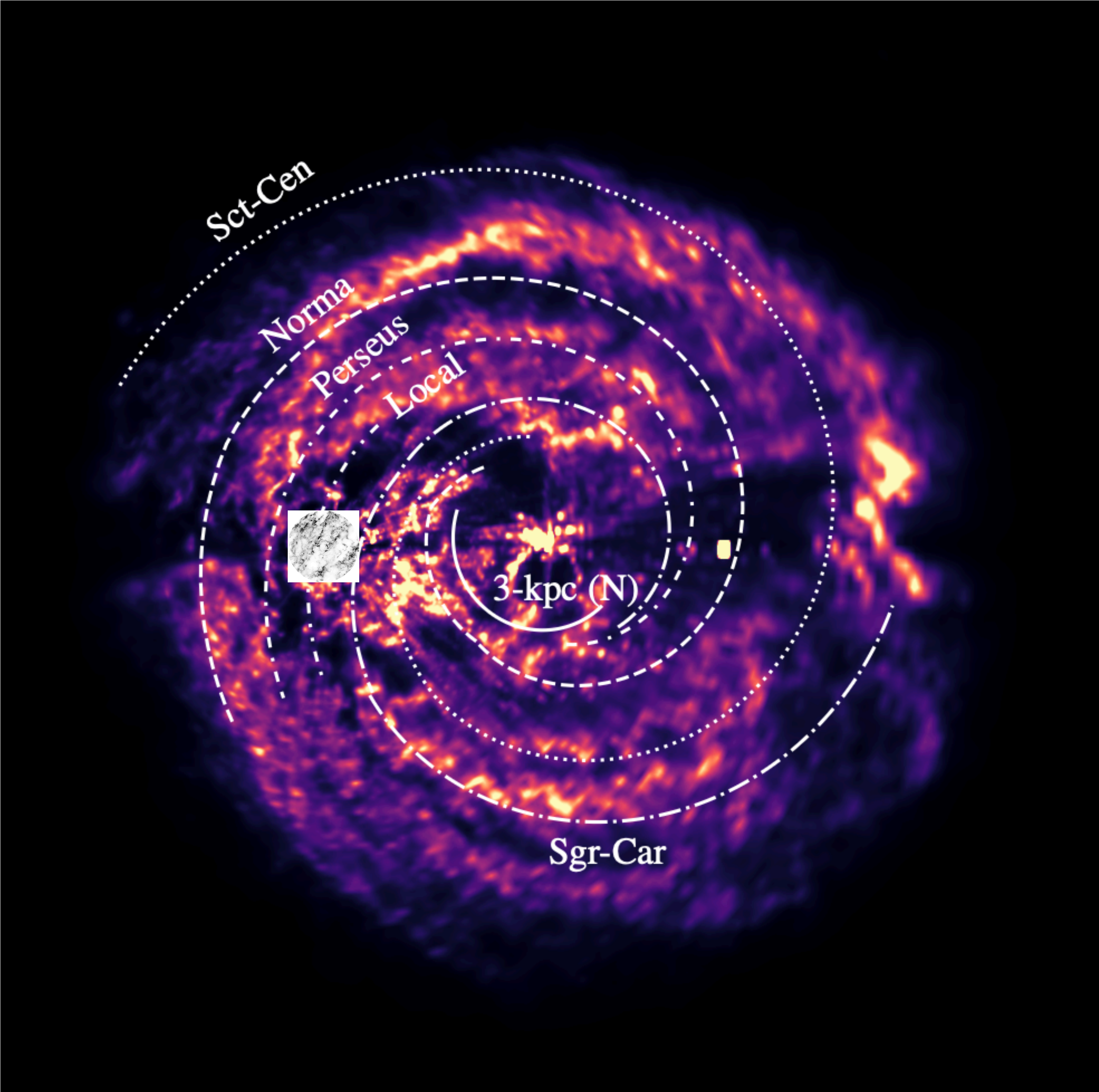


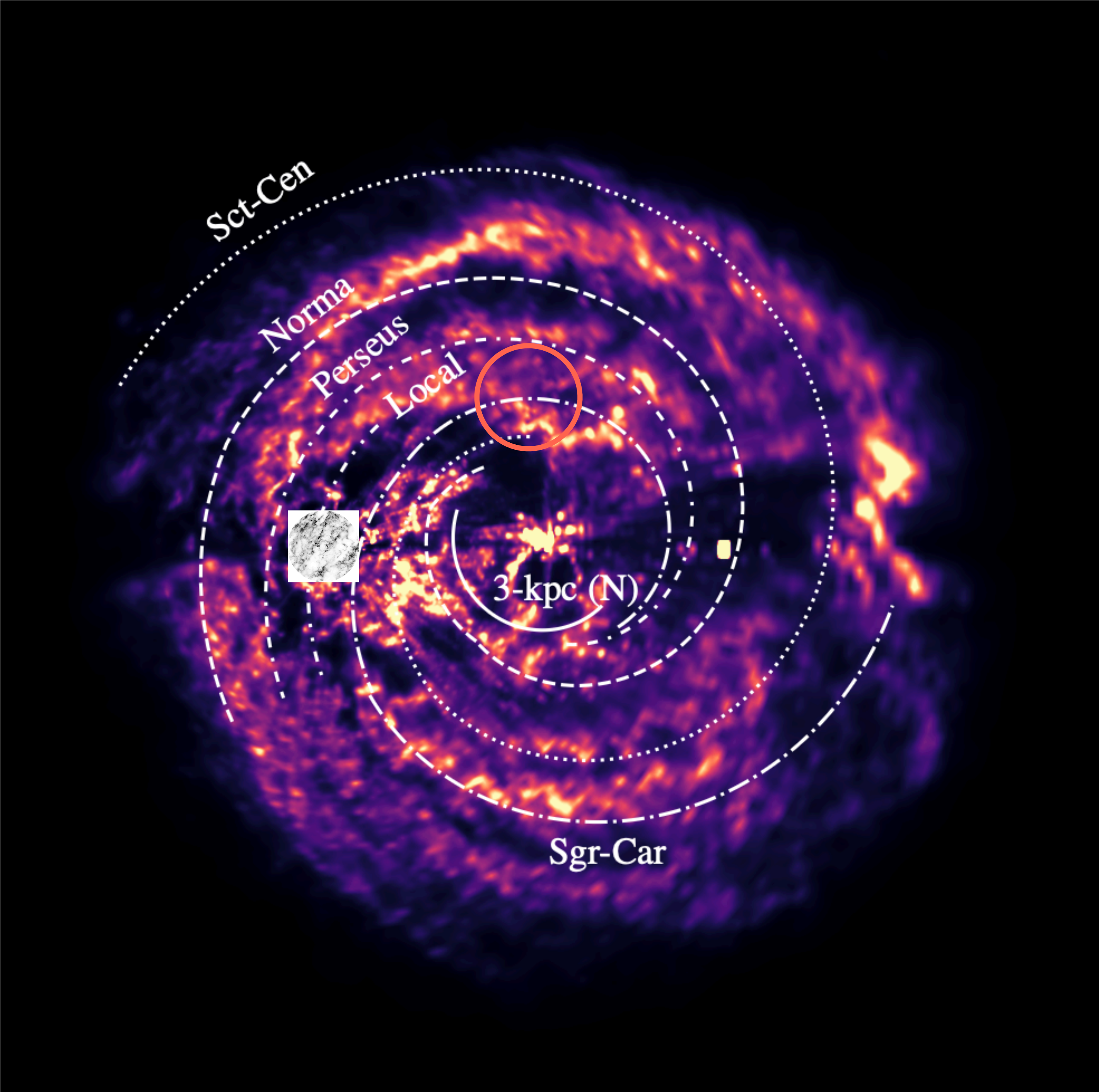


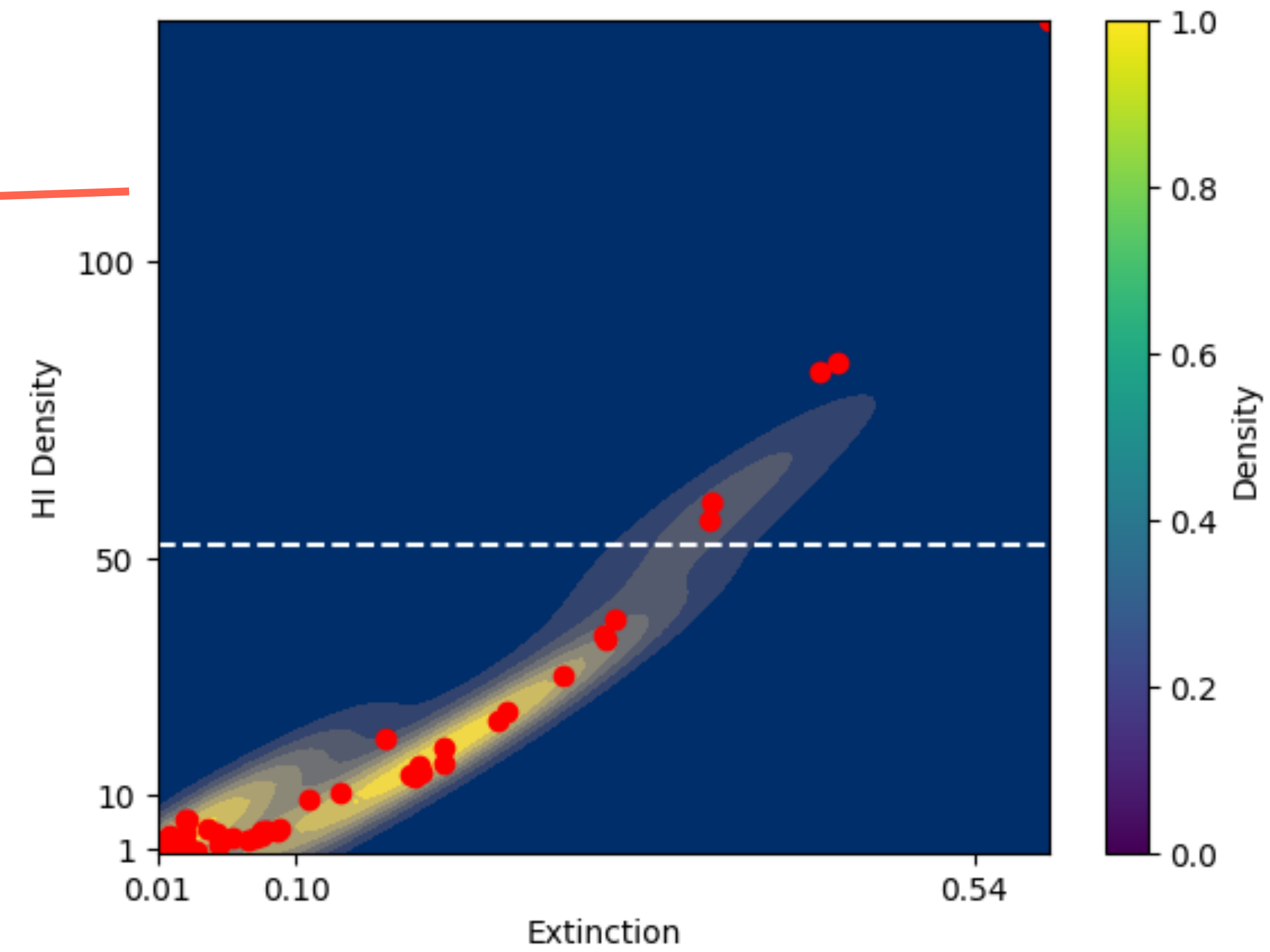
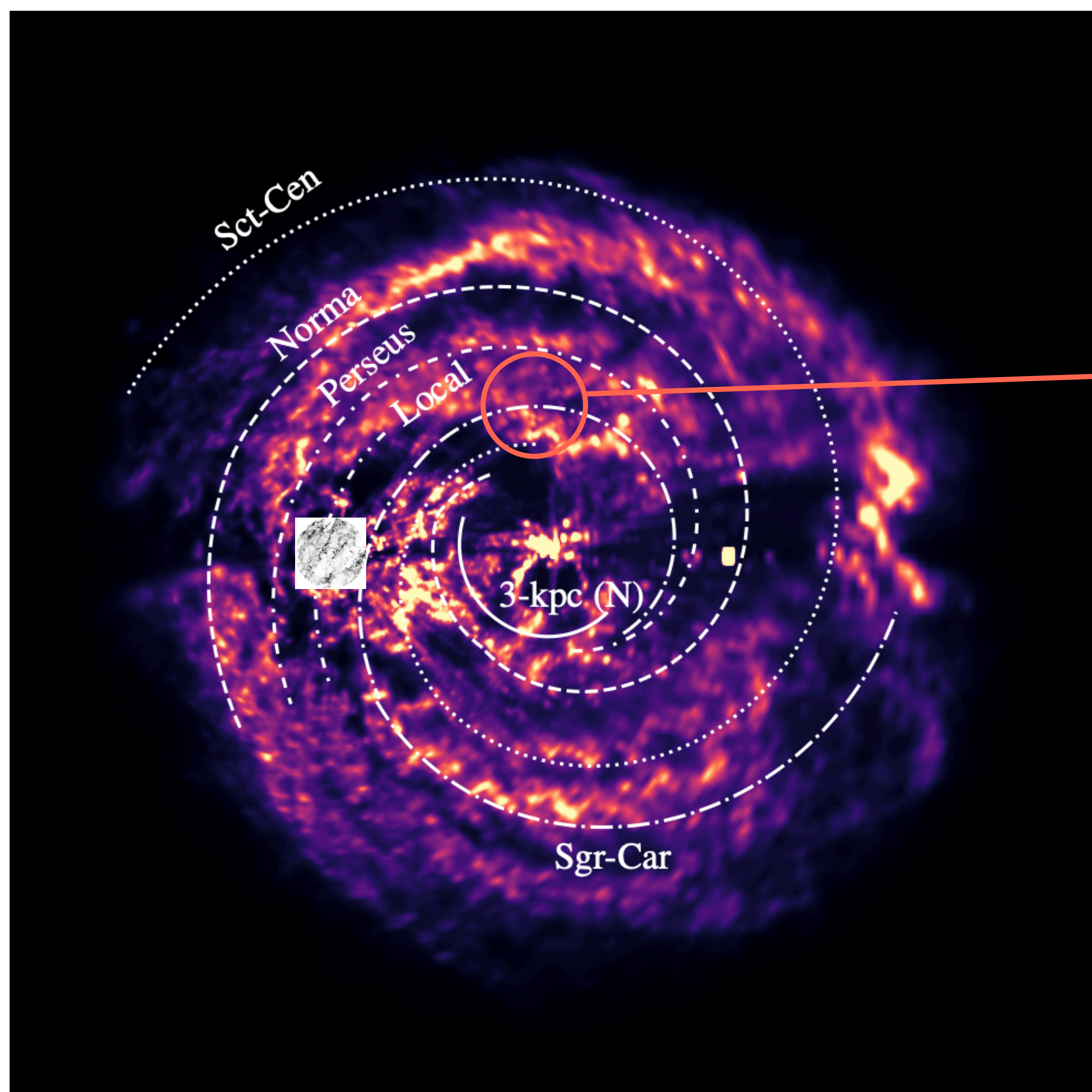


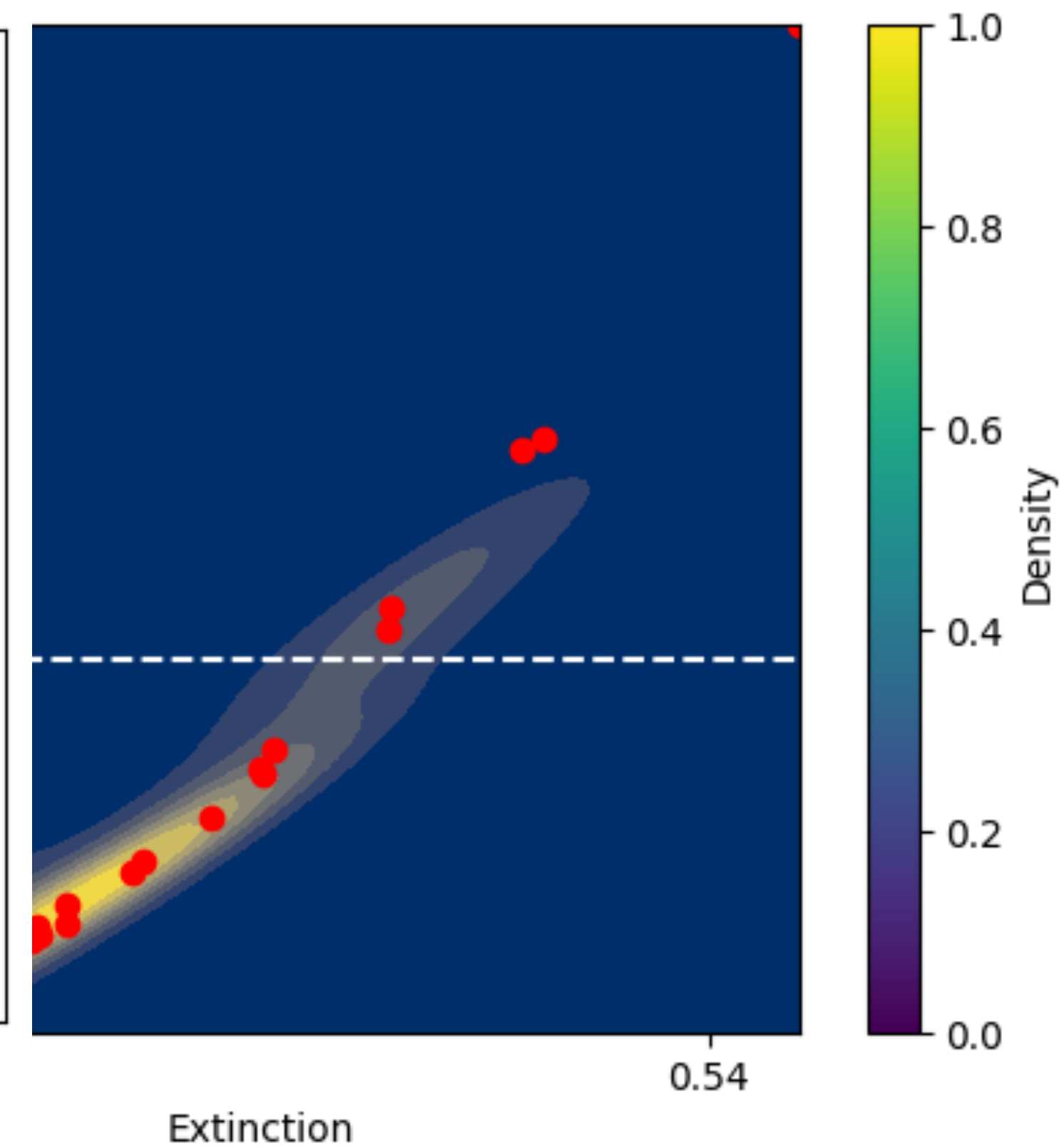
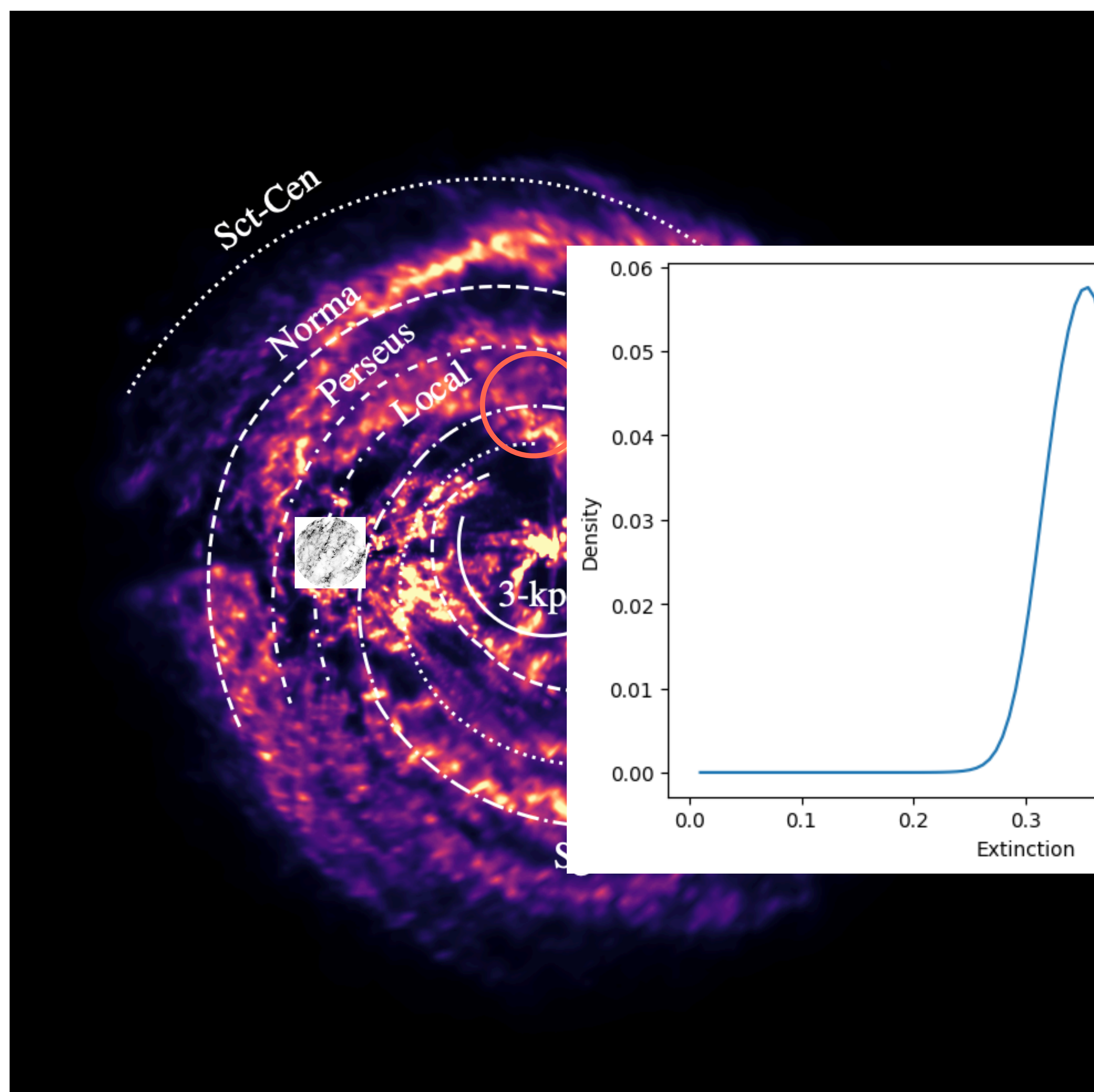


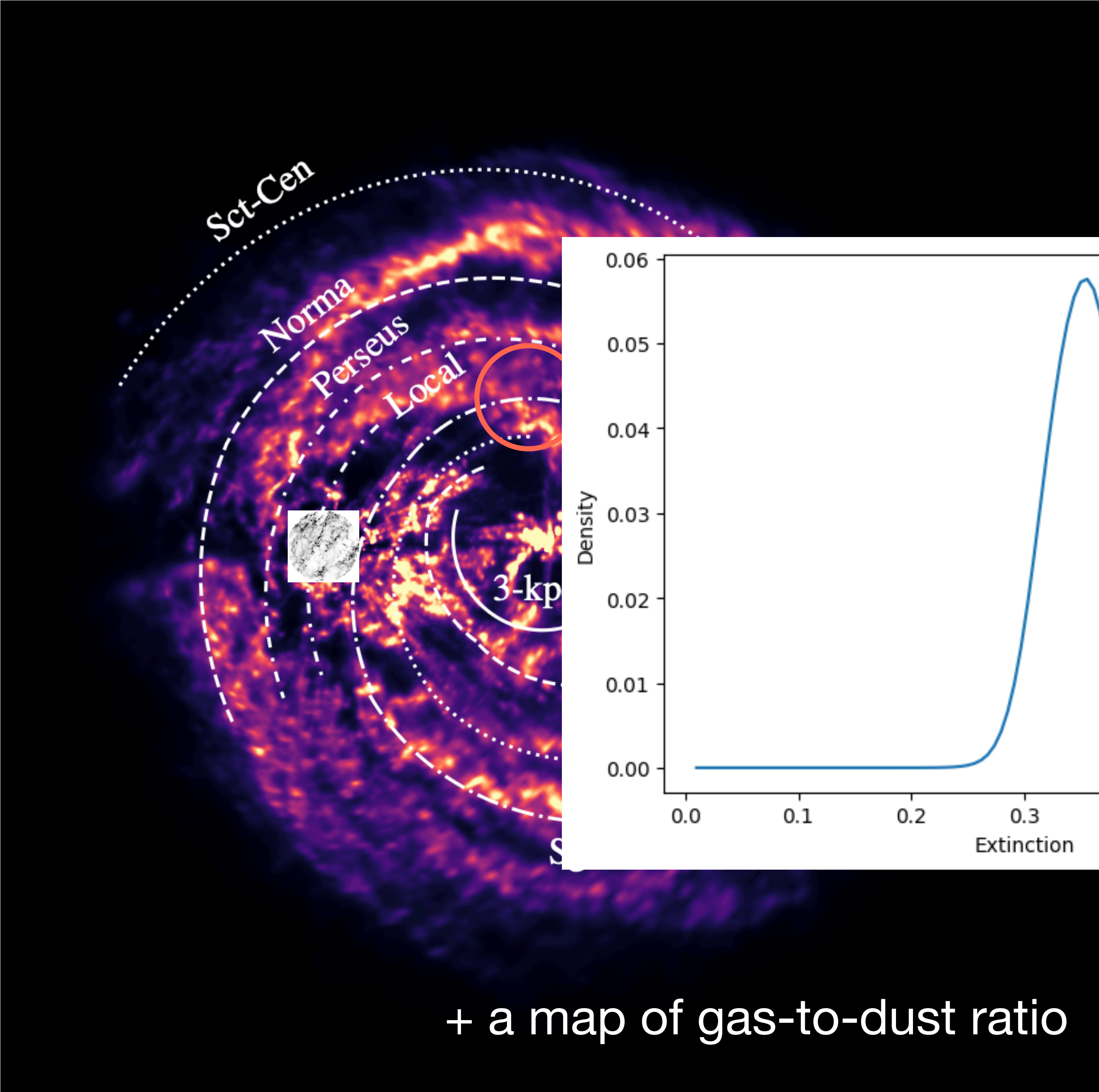








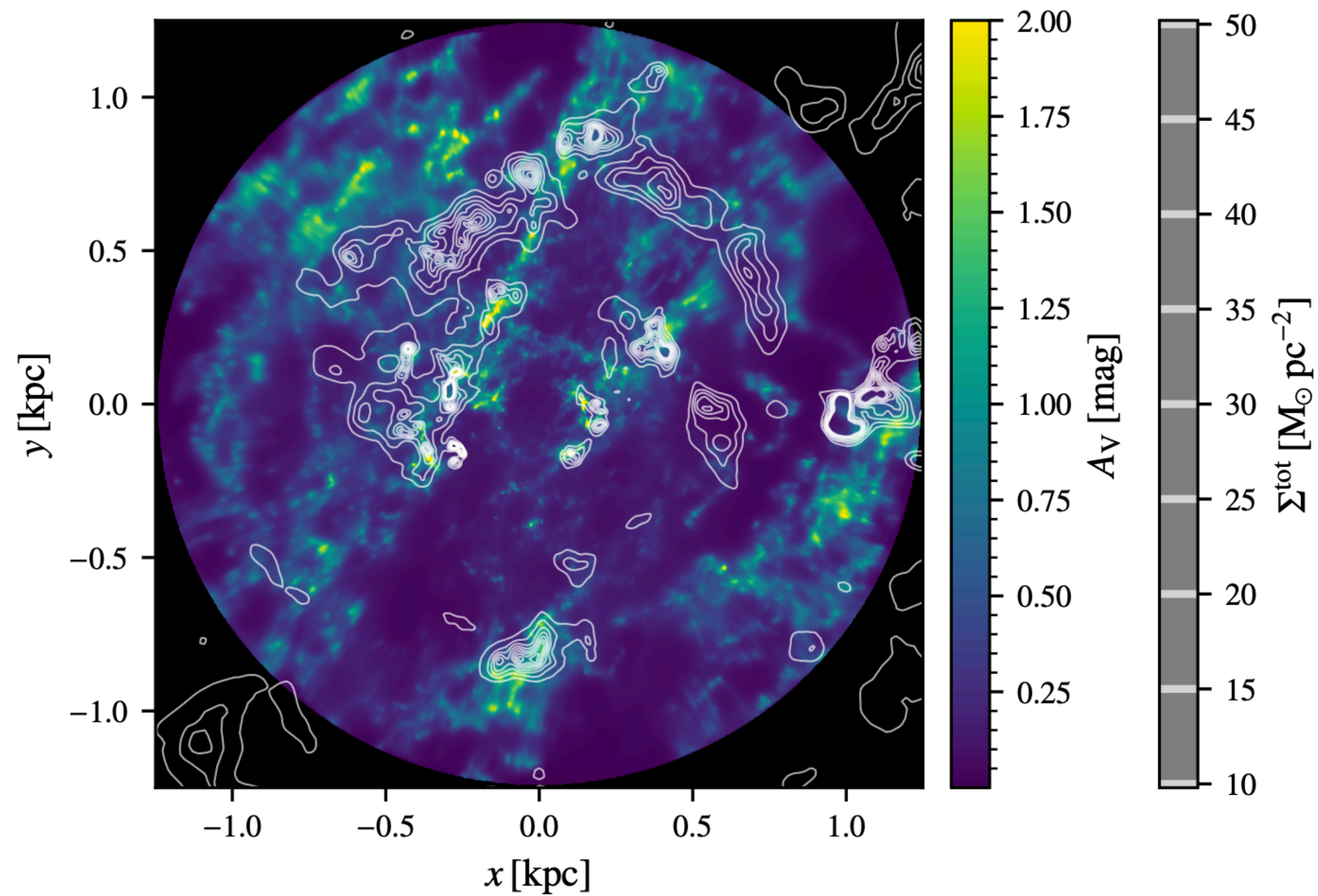




+ a map of gas-to-dust ratio

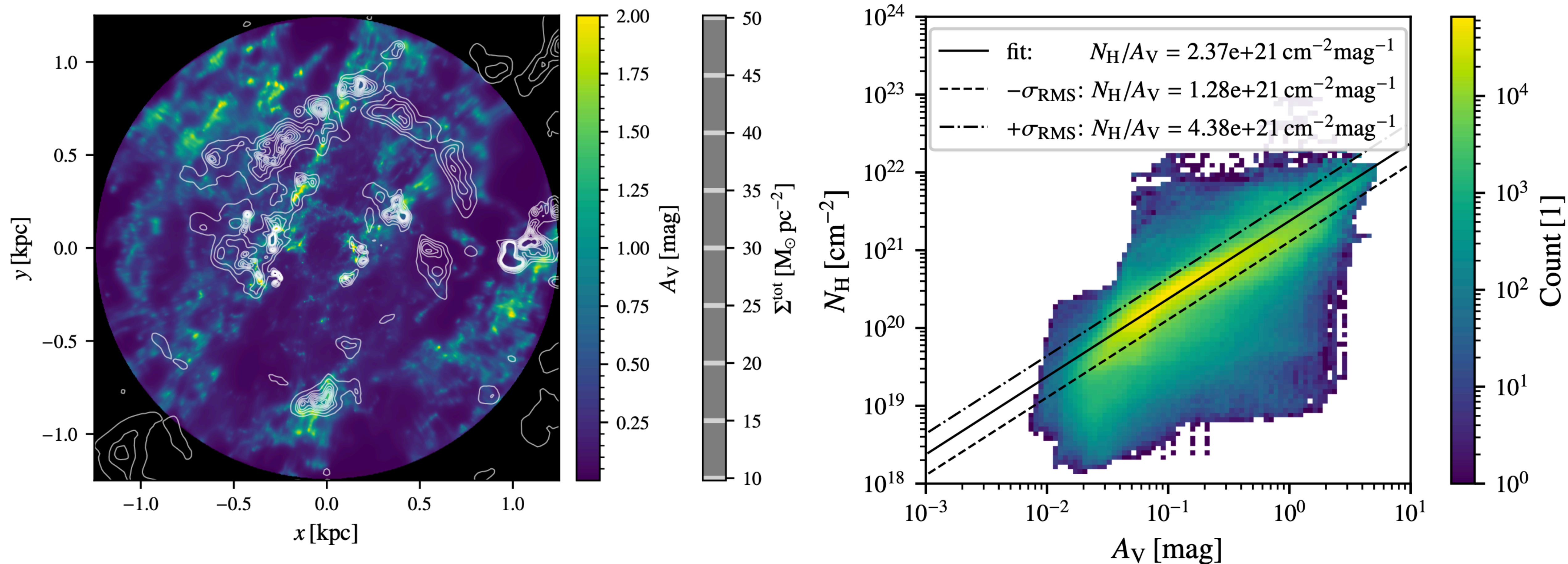
Inconsistencies between maps

Söding et al. 2024



Inconsistencies between maps

Söding et al. 2024



**How „accurate“ is the gas
inference?**

New velocity model

- Allow deviations of circular motions

Work In Progress

New velocity model

- Allow deviations of circular motions
- Achieved through first order expansion

Work In Progress

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$$r = r_C + \epsilon \qquad \theta = \theta_C + \delta$$

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$$\dot{r} = \dot{\epsilon} \qquad \dot{\theta} = \dot{\delta}$$

Work In Progress

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$$\dot{r} = \dot{\epsilon} \quad \dot{\theta} = \dot{\delta}$$

- Angular momentum conservation and equations of motions lead to

$$\epsilon = a \cdot \cos(n'(\phi - \phi_0))$$

Work In Progress

Current gas reconstruction

Work In Progress

How to test the accuracy?

- Mock test using simulated galaxies

Work In Progress

How to test the accuracy?

- Mock test using simulated galaxies
- Create mock observations of the simulated galaxy

Work In Progress

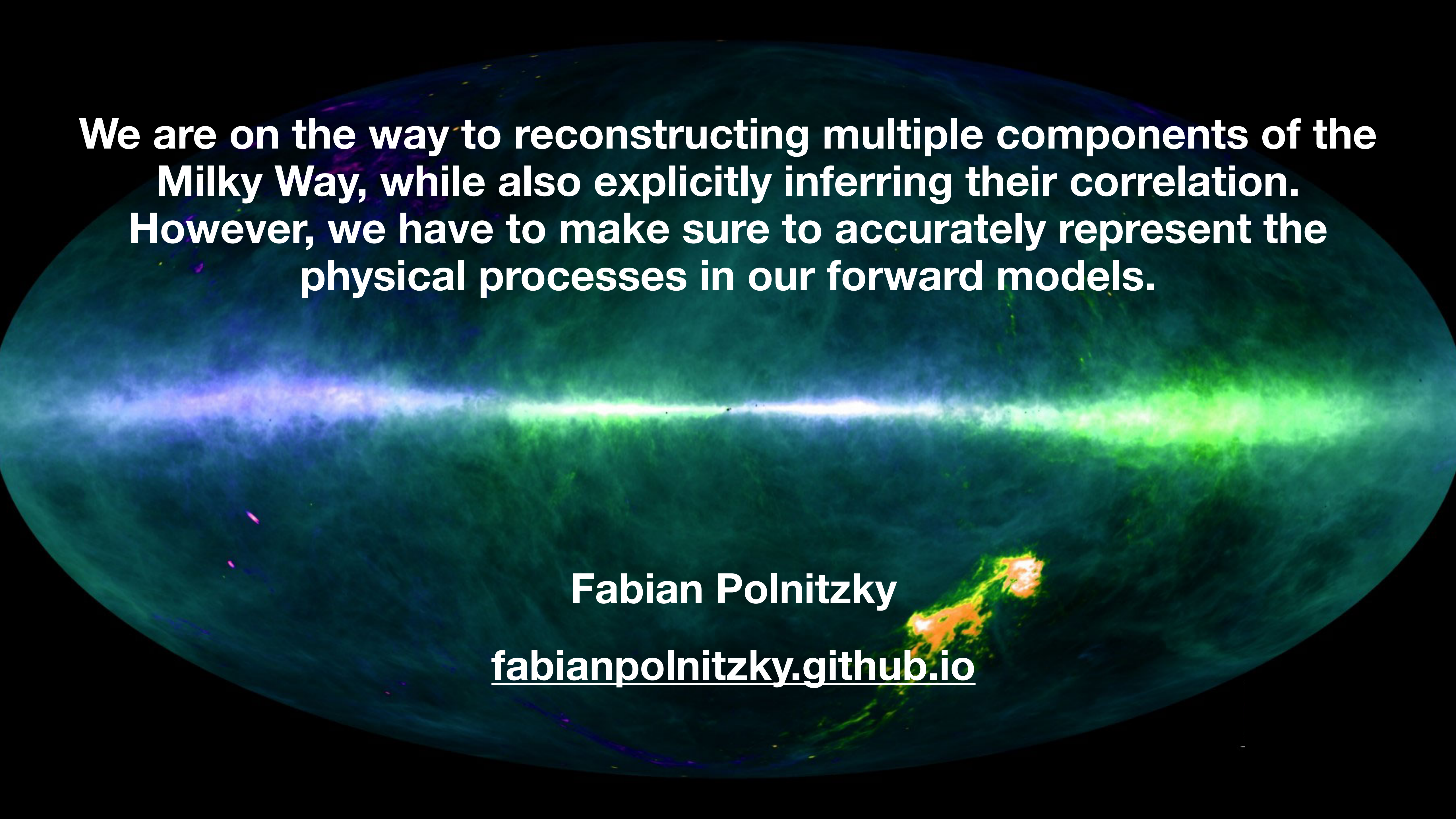
How to test the accuracy?

- Mock test using simulated galaxies
- Create mock observations of the simulated galaxy
- Infer the three-dimensional spatial distribution

Work In Progress

Mock observations

Work In Progress



We are on the way to reconstructing multiple components of the Milky Way, while also explicitly inferring their correlation. However, we have to make sure to accurately represent the physical processes in our forward models.

Fabian Polnitzky

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